

MSCCS 3.9



**M.Sc. Computer Science
Semester III**

(Revised Syllabus as Per NEP 2020)

**Data Visualization
Practical**

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Programme Name: M.Sc. Computer Science Semester III	Course Name: Data Visualization Practical
Total Credits: 02	Total Marks: 50
	University assessment: 50

Pre requisite: Knowledge of Power BI and Tableau

Course Outcome:

- Work with data analysis tools and perform data wrangling for practical purposes.
- Use Tableau to handle data from various sources and perform analysis of data.

Course Code	Course Title	Credits
PSCSP606b	Data Visualization Practical	02
Note: Following practical can be performed using Python and simulators, Raspberry Pi, and other hardware devices.		
1	Create Charts and Reports in Power BI.	
2	Time Intelligence and data analysis Functions with DAX	
3	Operations on Pinned Reports and Visuals using Power BI	
4	Create one-dimensional data using series and perform various operations on it	
5	Perform Reshaping of the hierarchical data and pivoting data frame data	
6	Connecting and extracting with various data resources in tableau and Perform calculations and creating parameters in Tableau.	
7	Designing Tableau Dashboards for different displays and devices	
8	Create a Trend model using data, Analyse-it and use it for forecasting.	
9	Creating Geospatial feature maps in Tableau using Geospatial Data.	
10	Create Dashboard and Storytelling using tableau.	

PRACTICAL 1

1

CREATE CHARTS & REPORTS IN POWER BI (WITH PYTHON- SIMULATED DATA)

Learning outcomes

- Generate realistic sample data using Python/simulators for BI practice
- Import, clean, and model data in **Power BI Desktop**
- Build core visuals (bar/line/map/treemap/table), add filters & slicers
- Write simple **DAX** measures (Revenue, Profit, YoY %, Return Rate)
- Design an executive report page and a drill-down analysis page
- (Optional) Use a **Python visual** inside Power BI

What you need

- Power BI Desktop (latest)
- The CSV above
- (Optional) Python installed locally for Power BI Python visuals

Dataset overview

The CSV contains ~8k retail transactions (Jan-2023 to Oct-2025) with fields like:

OrderDate, Region, Channel, CustomerSegment, Category, SubCategory, Product, Units, UnitPrice, Discount, Revenue, Cost, Profit, IsReturned, DeliveryDays.

Step-by-step

A. Import & quick model

1. Open **Power BI Desktop** → **Get Data** → **Text/CSV** → load the downloaded CSV.
2. In **Power Query**:
 - Ensure OrderDate is **Date** type.
 - Create a Month column: *Add Column* → *Date* → *Month* → *Name of Month* and a Year column.
 - Close & Apply.

3. (Optional best practice) Create a **Date table**

- *Model view* → **New table**:
- DateTable =
- `CALENDAR ( DATE(2023,1,1), DATE(2025,12,31) )`
- Add:
- `Year = YEAR(DateTable[Date])`
- `Month = FORMAT(DateTable[Date], "MMM")`
- `MonthNum = MONTH(DateTable[Date])`
- Mark as **Date table**. Relate DateTable[Date] → OrderDate.

B. Core DAX measures

Add a few measures (*Model view* → your table → New measure):

Total Revenue = `SUM('retail_sales_powerbi_practice'[Revenue])`

Total Cost = `SUM('retail_sales_powerbi_practice'[Cost])`

Total Profit = `[Total Revenue] - [Total Cost]`

Profit % = `DIVIDE([Total Profit], [Total Revenue])`

Orders =
`DISTINCTCOUNT('retail_sales_powerbi_practice'[OrderID])`

Return Rate % =
`DIVIDE(SUM('retail_sales_powerbi_practice'[IsReturned]), [Orders])`

YoY Revenue % =

VAR ThisYear =

`CALCULATE([Total Revenue], ALL('DateTable'[Year]),
VALUES('DateTable'[Year]))`

VAR PrevYear =

`CALCULATE([Total Revenue], ALL('DateTable'[Year]),
VALUES('DateTable'[Year]) - 1)`

`RETURN DIVIDE(ThisYear - PrevYear, PrevYear)`

C. Build visuals (Report view)

Page 1 – Executive overview

- **Cards:** Total Revenue, Total Profit, Profit %, Orders, Return Rate %.

- **Line chart:** Revenue by DateTable[MonthNum] (axis) & Year (legend).
- **Column chart:** Profit by Region.
- **Treemap:** Revenue by Category → SubCategory.
- **Table:** Product, Units, Revenue, Profit, Profit %.
- **Slicers:** Year, Region, Channel, CustomerSegment.

Page 2 – Drill-down analysis

- **Stacked column:** Revenue by Channel with CustomerSegment as legend.
- **Clustered bar:** Profit by Product (Top N filter = 15).
- **Line:** Return Rate % by Month (use Date hierarchy).
- **Decomposition Tree** (if enabled): Analyze Profit by Region → Category → Product.

Formatting tips

- Use a consistent theme (View → Themes).
- Align/Distribute for neat layout; add subtitles and data labels where helpful.
- Add a **tooltip page** (Mini report with KPIs) and set as tooltip for multiple visuals.

D. (Optional) Python visual inside Power BI

- File → Options → **Python scripting** → select your Python home.
- Insert a **Python visual**, drag fields (e.g., OrderDate, Revenue, Region) to the Values well, then use a script like:

```
import pandas as pd

from matplotlib import pyplot as plt

df = dataset.copy()

df['OrderDate'] = pd.to_datetime(df['OrderDate'])

m = df.groupby([df['OrderDate'].dt.to_period('M').astype(str),
'Region'])['Revenue'].sum().reset_index()

pivot = m.pivot(index='OrderDate', columns='Region',
values='Revenue').fillna(0)

plt.figure()

pivot.plot() # Power BI runs this in its own environment
```

```
plt.title('Monthly Revenue by Region')
```

```
plt.xlabel('Month')
```

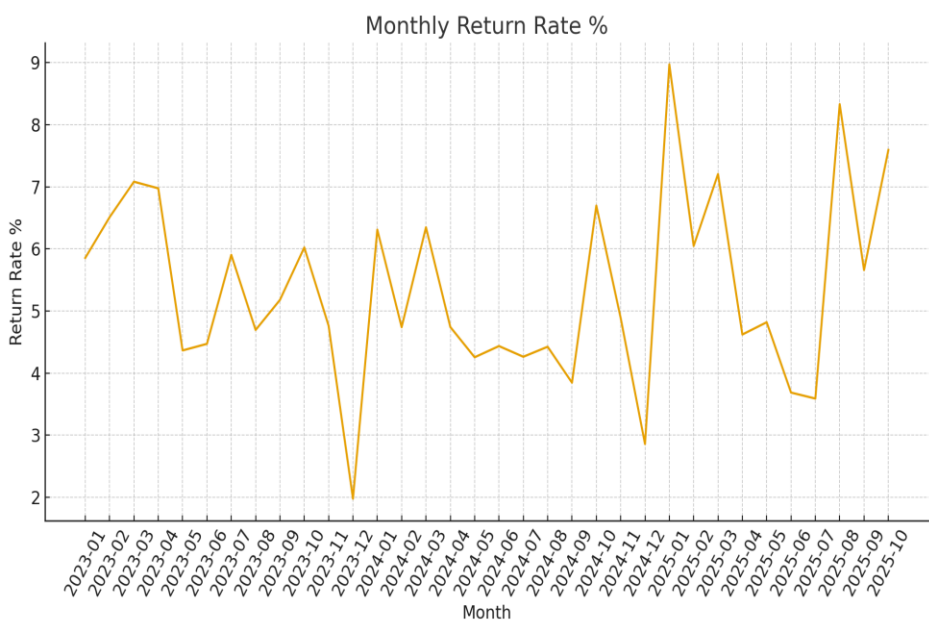
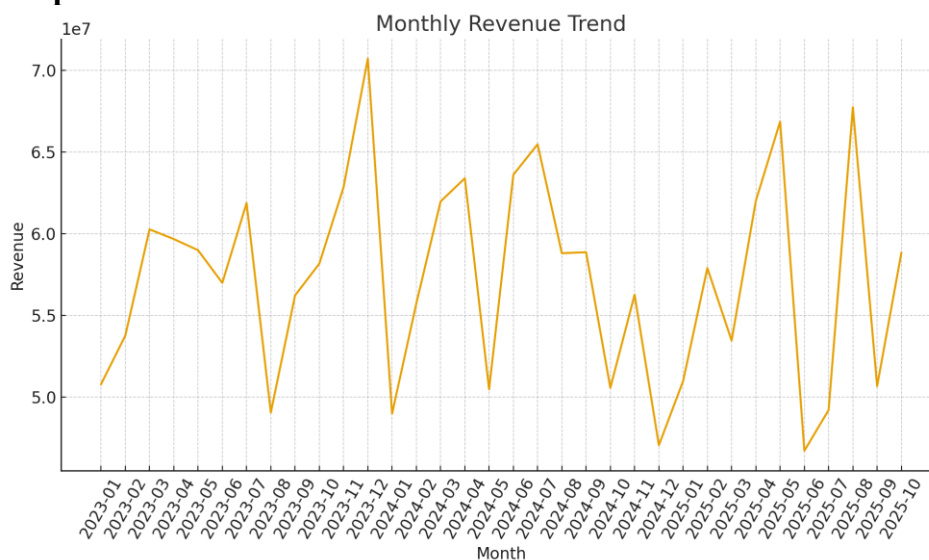
```
plt.ylabel('Revenue')
```

```
plt.tight_layout()
```

E. Publish & share (if using Power BI Service)

- Sign in → **Publish** to a workspace.
- Pin key visuals to a **dashboard**; configure alerts on KPIs (e.g., Profit % < 10%).

Output



Viva / reflection questions

Create Charts &
Reports in Power BI
(With Python-
Simulated Data)

1. Why use a dedicated Date table, and how does it improve time intelligence?
2. Difference between a **calculated column** and a **measure** in DAX with an example.
3. When would you prefer a **treemap** vs a **stacked bar**?
4. How do slicers interact with the filter context for measures like Profit %?
5. One risk of using Python visuals in production dashboards?



TIME INTELLIGENCE & DATA ANALYSIS FUNCTIONS WITH DAX

Learning outcomes :

- Build and mark a **Date table**.
- Create **YTD, QTD, MTD, YoY, MoM** measures in DAX.
- Interpret trends via **line, bar, and KPI** visuals.

What you need

- Power BI Desktop
- The CSV above

Dataset fields (same as P1)

OrderDate, Region, Channel, CustomerSegment, Category, SubCategory, Product, Units, UnitPrice, Discount, Revenue, Cost, Profit, IsReturned, DeliveryDays

Step-by-step (in Power BI)

1. Create Date table

```
DataTable = CALENDAR ( DATE(2023,1,1), DATE(2025,12,31) )
```

```
Year = YEAR ( DataTable[Date] )
```

```
Month = FORMAT ( DataTable[Date], "MMM" )
```

```
Quarter = "Q" & FORMAT ( DataTable[Date], "Q" )
```

Mark as Date table; relate DataTable[Date] → fact [OrderDate].

2. Base measures

```
Total Revenue = SUM('retail_sales_powerbi_practice'[Revenue])
```

```
Total Profit = SUM('retail_sales_powerbi_practice'[Profit])
```

3. Time-intelligence measures

```
Revenue YTD = TOTALYTD([Total Revenue], 'DataTable'[Date])
```

```
Revenue QTD = TOTALQTD([Total Revenue], 'DataTable'[Date])
```

```
Revenue MTD = TOTALMTD([Total Revenue], 'DataTable'[Date])
```

Revenue PY = CALCULATE([Total Revenue],
SAMEPERIODLASTYEAR('DateTable'[Date]))

YoY % = DIVIDE([Total Revenue] - [Revenue PY], [Revenue PY])

Revenue PM = CALCULATE([Total Revenue],
PREVIOUSMONTH('DateTable'[Date]))

MoM % = DIVIDE([Total Revenue] - [Revenue PM], [Revenue PM])

Running Total =

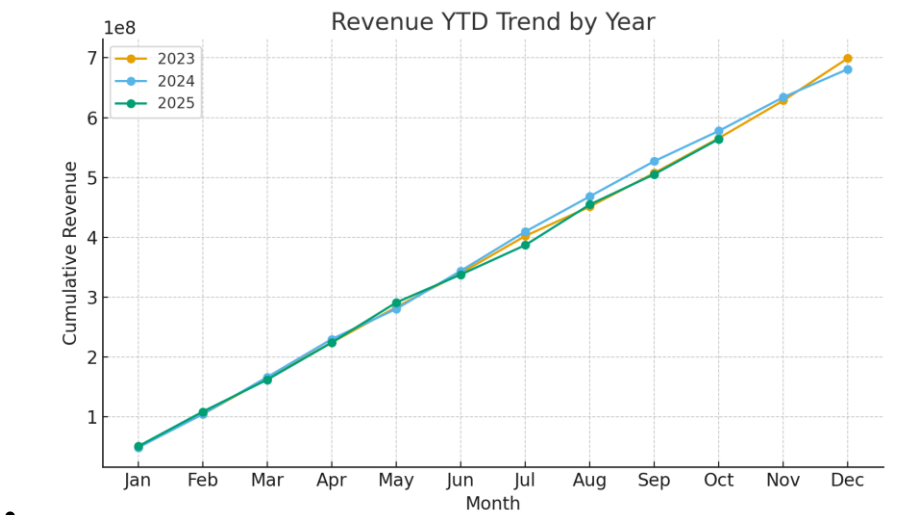
CALCULATE([Total Revenue],
FILTER(ALL('DateTable'[Date]), 'DateTable'[Date] <=
MAX('DateTable'[Date])))
)

4. Report layout

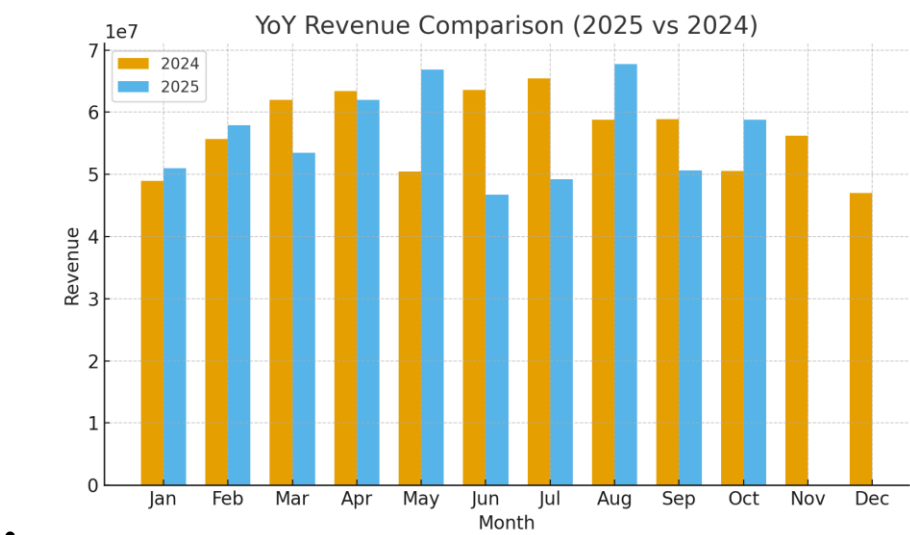
- **Page 1 – Trend:** Line chart → Revenue YTD by Date (legend: Year); KPI: MoM %.
- **Page 2 – Comparison:** Clustered bars → Total Revenue by Month (2025 vs 2024); Line/Bar combo → Total Revenue + Running Total.
- **Table:** Month, Total Revenue, Revenue PY, YoY %, MoM % for sanity checks.
- Slicers: Year, Region, Channel.

Output (simulated with Python)

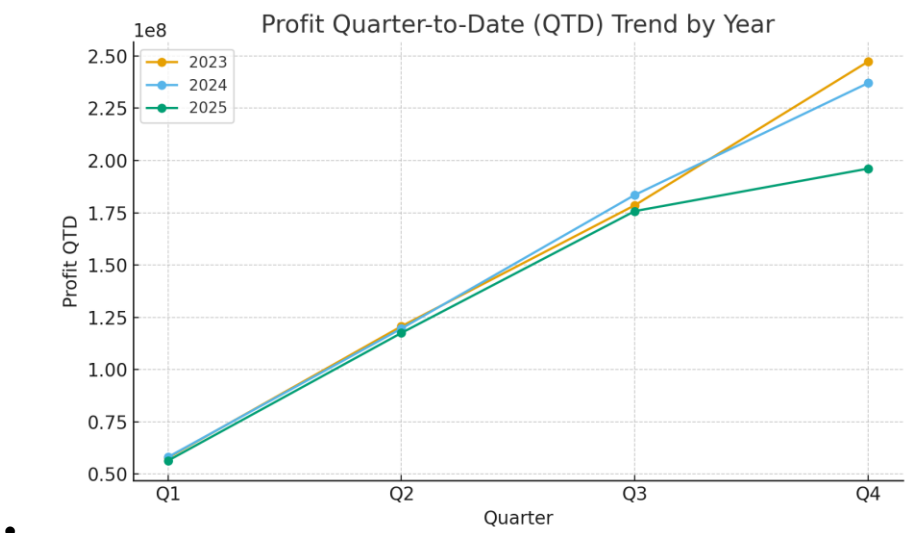
- **Summary table** (recent months): shown in the UI as “*Practical 2 – Time Intelligence Summary (Recent 12 months)*”.
- **Charts rendered here:**
 1. Revenue **YTD** trend by year
 2. **YoY** revenue comparison (2025 vs 2024)
 3. Profit **QTD** trend by year



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PRACTICAL 3

3

OPERATIONS ON PINNED REPORTS AND VISUALS USING POWER BI

Learning Outcomes

- Understand the difference between **Reports**, **Dashboards**, and **Tiles**.
- Learn how to **pin visuals** and **manage pinned content** in Power BI Service.
- Perform **refresh**, **delete**, **duplicate**, and **rearrange** operations on pinned visuals.
- Use **interactive dashboard features**: drill-through, Q&A, and alerts.

Software and Resources

Requirement Details

Application	Power BI Desktop + Power BI Service (cloud)
Dataset	retail_sales_powerbi_practice.csv (<i>from Practical 1</i>)
Account	Power BI (organizational or student account)
Prerequisite	Report created in Practical 1 and 2 containing Revenue and Profit visuals

Concept Overview

- **Report:** A multi-page canvas built in Power BI Desktop.
- **Dashboard:** A single-page collection of **pinned visuals** (tiles) created in Power BI Service.
- **Pinning:** The process of sending a visual (chart, KPI, card) from a report to a dashboard.
- **Tile:** The interactive card on a dashboard that displays pinned content.

Power BI allows **live tiles** (connected to the dataset) that automatically refresh when the underlying data changes.

Step-by-Step Procedure

Publish the Report

1. Open your Practical 2 Power BI report.

2. Click **Publish** → **Select Workspace** (e.g., *My Workspace*).
3. After publishing, open **app.powerbi.com** → navigate to your workspace → open the report.

Pin Visuals to a Dashboard

1. Hover over a visual (e.g., *Revenue YTD line chart*).
2. Click the  **Pin Visual** icon.
3. Choose **Existing Dashboard** (or create a new one, e.g., *Retail Performance Board*).
4. Repeat for:
 - KPI cards (Total Revenue, Profit %, Return Rate)
 - Trend charts (YTD and QTD)
 - YoY bar comparison

Each pinned element now appears as a **tile** on your dashboard.

Perform Operations on Pinned Visuals

Operation	Steps
Move / Resize	Drag corners or edges of a tile to resize or re-order.
Duplicate	Click “More options (⋮)” → “Duplicate tile”.
Refresh	Dashboard → Refresh button or Schedule Refresh from Dataset settings.
Delete	“More options (⋮)” → “Delete tile”.
Edit Details	Change title, subtitle, background color under Tile details.
Pin Live Page	Use “Pin a Live Page” to keep interactivity from entire report page.
Drill through Tile	Click a tile → Open report for detailed view.
Alerts	For KPI tiles, set “Manage alerts” to notify when value crosses threshold.

Create Dashboard Interactions

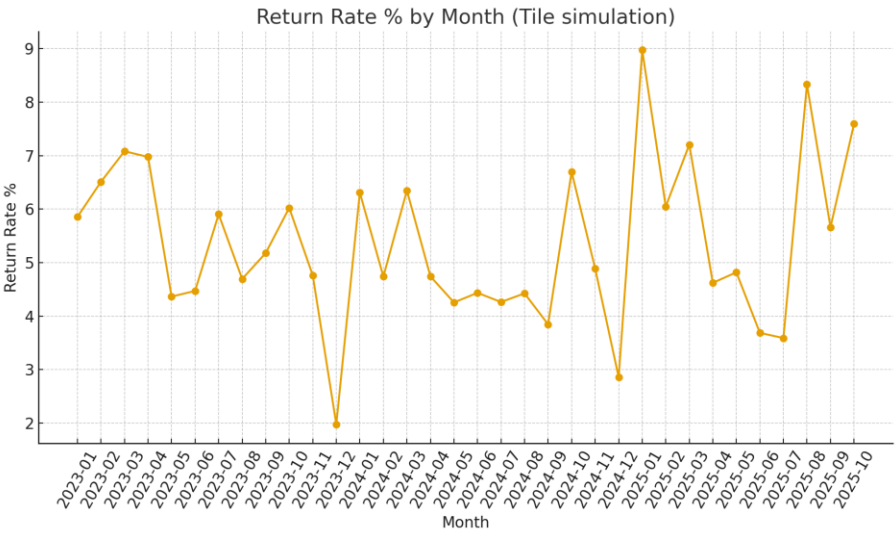
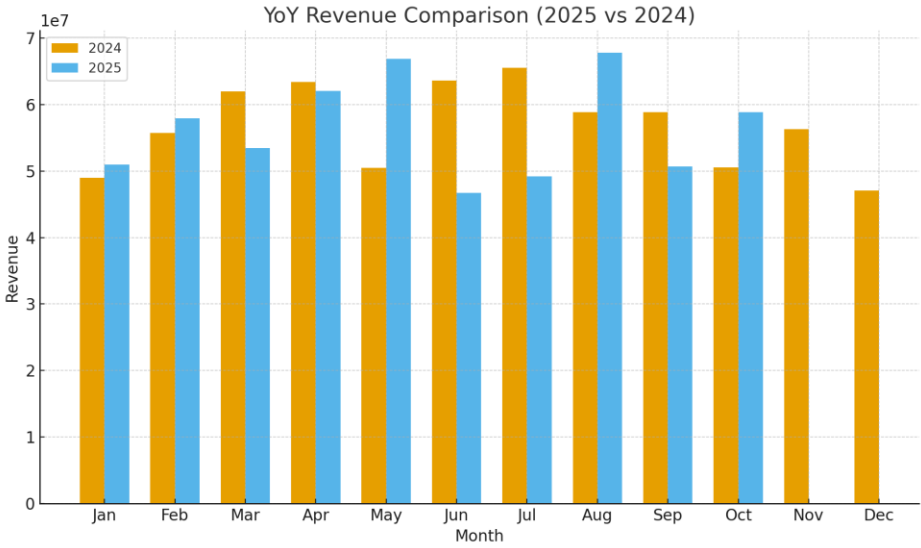
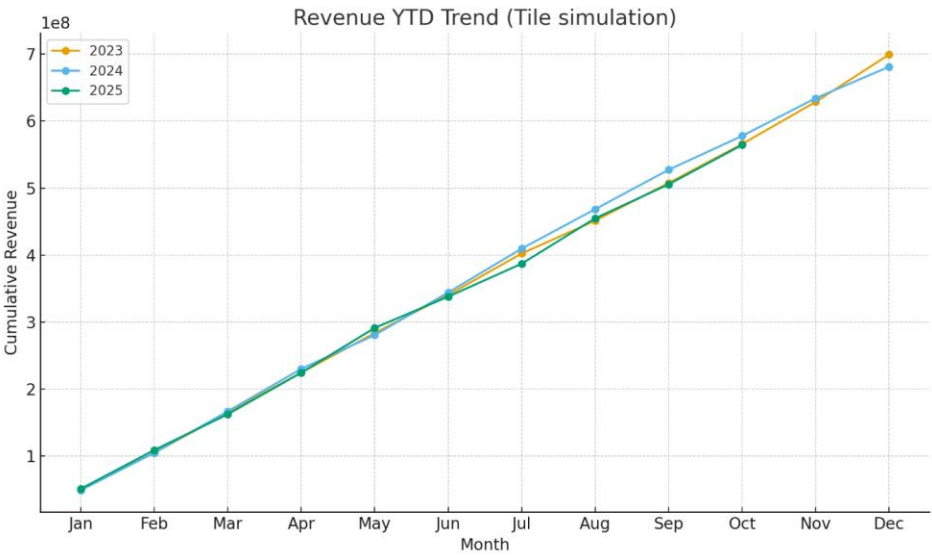
Operations on Pinned Reports and Visuals using Power BI

- 1. Add **text boxes**, **images**, and **buttons** for branding (VSIT logo, semester title).
- 2. Add **web link** tiles to open college portal or Google Sheet summary.
- 3. Use **Pin from Q&A** → Ask natural language query (e.g., “show revenue by region in 2025”) → pin the result.

Expected Dashboard Layout

Tile	Visual Type	Purpose
Total Revenue	KPI Card	Shows real-time Revenue with YoY trend
Profit %	KPI Card	Highlights profitability indicator
Revenue YTD	Line Chart	Monthly cumulative trend
YoY Comparison	Clustered Bar	2024 vs 2025 revenue comparison
Profit QTD	Line Chart	Quarterly accumulation
Return Rate	Gauge Card or	Performance quality indicator

Pinned Dashboard – KPI Tiles (Simulated)			
	KPI	Value	
1	Total Revenue	1944840461.16	
2	Total Profit	680429739.47	
3	Profit %	34.99	
4	Orders	8000.0	
5	Return Rate %	5.38	



PRACTICAL 4

4

CREATE ONE-DIMENSIONAL DATA USING SERIES AND PERFORM VARIOUS OPERATIONS ON IT

Learning Outcomes

- Understand **Pandas Series** – the one-dimensional labeled array in Python.
- Learn to **create Series** using lists, NumPy arrays, and dictionaries.
- Perform **indexing, slicing, arithmetic, statistical, and filtering operations**.
- Visualize the Series data using **Matplotlib**.

Software and Libraries Required

Tool	Purpose
------	---------

Python 3.x	Programming environment
------------	-------------------------

Pandas	For Series creation and data analysis
--------	---------------------------------------

NumPy	For numerical operations
-------	--------------------------

Matplotlib	For visualization
------------	-------------------

Concept Overview

- A **Series** is a **one-dimensional labeled array** capable of holding any data type.
- It consists of two components:
 - **Data values** (the actual elements)
 - **Index labels** (used for identification)
- Think of it as an **Excel column** or **dictionary with ordered keys**.

Python Code

```
# Practical 4: Create One-Dimensional Data using Series and perform operations
```

```
import pandas as pd
```

```
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Create a Series of Weekly Sales
sales = pd.Series([23000, 17000, 25000, 30000, 28000, 32000, 21000],
                  index=['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun'],
                  name="Weekly Sales")

print("Original Series:\n", sales)

# Step 2: Perform basic operations
print("\n--- Summary Statistics ---")
print("Mean:", sales.mean())
print("Maximum:", sales.max())
print("Minimum:", sales.min())
print("Standard Deviation:", round(sales.std(), 2))

# Step 3: Element-wise operations
sales_10 = sales * 1.10 # Increase by 10%
print("\nSales Increased by 10%:\n", sales_10)

# Step 4: Conditional filtering
print("\nDays with High Sales (>25000):\n", sales[sales > 25000])

# Step 5: Indexing and sorting
print("\nSorted Sales (Descending):\n",
      sales.sort_values(ascending=False))

print("\nSelected Days (Mon, Fri, Sun):\n", sales.loc[['Mon', 'Fri', 'Sun']])

# Step 6: Visualization
sales.plot(kind='bar', title="Weekly Sales", ylabel="Amount",
           xlabel="Day")

plt.show()

sales_10.plot(kind='line', marker='o', title="Sales Increased by 10%",
              ylabel="Amount", xlabel="Day")

plt.show()
```

Simulated Output (Python Execution)

Original Series

Day Sales

Mon 23000

Tue 17000

Wed 25000

Thu 30000

Fri 28000

Sat 32000

Sun 21000

Summary Statistics

Operation	Value
Mean	25142.86
Maximum	32000
Minimum	17000

Standard Deviation 5165.25

Sales Increased by 10%

Day Sales (Increased)

Mon 25300

Tue 18700

Wed 27500

Thu 33000

Fri 30800

Sat 35200

Sun 23100

Create One-Dimensional Data using Series and Perform Various Operations on It

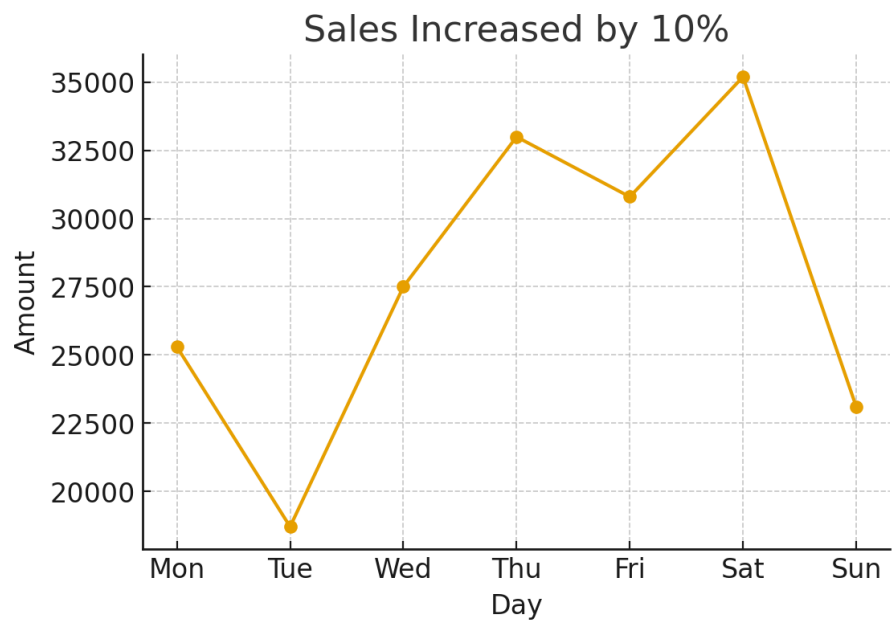
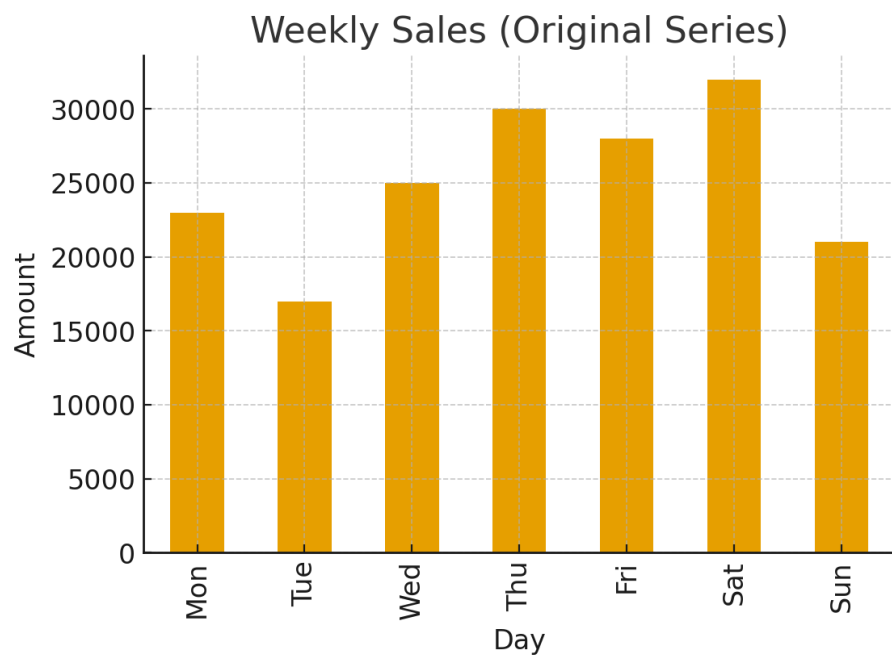
 High Sales Days (>25000)

Day Sales

Thu 30000

Fri 28000

Sat 32000



PRACTICAL 5

5

RESHAPING OF HIERARCHICAL DATA AND PIVOTING DATAFRAME DATA

Learning Outcomes

- Understand **hierarchical (multi-index) data representation** in Pandas.
- Learn how to **reshape data** between long and wide formats.
- Apply **pivot, pivot_table, stack, unstack, melt, and crosstab** operations.
- Visualize reshaped data through bar and heatmap-like plots.

Step-by-Step Implementation

Step 1 – Import Libraries & Create Hierarchical Data

```
import pandas as pd, numpy as np, matplotlib.pyplot as plt

rng = np.random.default_rng(7)

dates = pd.date_range("2025-01-01", periods=12, freq="M")
regions = ["North", "South", "East", "West"]
stores = [f"S{i:02d}" for i in range(1,7)]
categories = ["Electronics", "Furniture", "Office Supplies"]
rows=[]

for d in dates:
    for r in regions:
        for s in rng.choice(stores, size=3, replace=False):
            for c in categories:
                units = rng.integers(5,80)

                price = rng.uniform(200,5000) if c!="Office Supplies" else
rng.uniform(20,600)

                revenue = round(units*price,2)

                rows.append([d,r,s,c,units,round(price,2),revenue])
```

```
df = pd.DataFrame(rows,
columns=["Date", "Region", "Store", "Category", "Units", "UnitPrice", "Revenue"])
```

Creates 12-month data for 4 regions × 3 stores × 3 categories.

Step 2 – Set a Hierarchical Index

```
midx = df.set_index(["Region", "Store", "Date", "Category"]).sort_index()
```

Now the DataFrame uses **Region > Store > Date > Category** as hierarchy.

Step 3 – Reshape Using Stack / Unstack

```
unstacked = midx["Revenue"].unstack("Category") # Wide
```

```
restacked = unstacked.stack().to_frame(name="Revenue") # Back to long
```

- **Unstack:** moves “Category” level to columns.
- **Stack:** reverses it, restoring to the original long form.

Step 4 – Pivot and Pivot Table

```
sample =
df[df["Region"]=="North"].assign(Month=df["Date"].dt.strftime("%b"))
```

```
pivot = pd.pivot_table(sample, index="Month", columns="Category",
                        values="Revenue", aggfunc="sum")
```

→ Creates a **Month × Category** matrix for the North region.

```
pt = pd.pivot_table(df.assign(Month=df["Date"].dt.strftime("%b")),
                    values="Revenue", index=["Region", "Month"],
                    columns="Category", aggfunc="sum",
                    margins=True, margins_name="Total").round(2)
```

→ Summarizes revenue across regions and months (with **Total** row / column).

Step 5 – Swap Levels & Sort

```
swapped = midx.swaplevel("Store", "Region").sort_index()
```

Switches hierarchy order between Region and Store.

Step 6 – Melt (Wide → Long)

```
wide = df.pivot_table(values="Revenue",
index=["Region", "Store", "Date"],
                        columns="Category", aggfunc="sum").reset_index()
```

```
long = wide.melt(id_vars=["Region","Store","Date"],
                 value_vars=categories,
                 var_name="Category", value_name="Revenue")
```

Converts back to long format suitable for analytics or visualization.

Step 7 – Crosstab

```
ct = pd.crosstab(df["Region"], df["Category"],
                 values=df["Units"], aggfunc="sum").fillna(0).astype(int)
```

Creates a **Region × Category** summary of total units sold.

Step 8 – Visualization

```
sum_by_region = df.groupby(["Region","Category"])["Revenue"].sum().unstack()

sum_by_region.plot(kind="bar", title="Revenue by Region & Category")

plt.tight_layout(); plt.show()

# Heatmap-like view

plt.imshow(pivot.fillna(0).values, aspect="auto")

plt.title("North Region – Monthly Revenue")

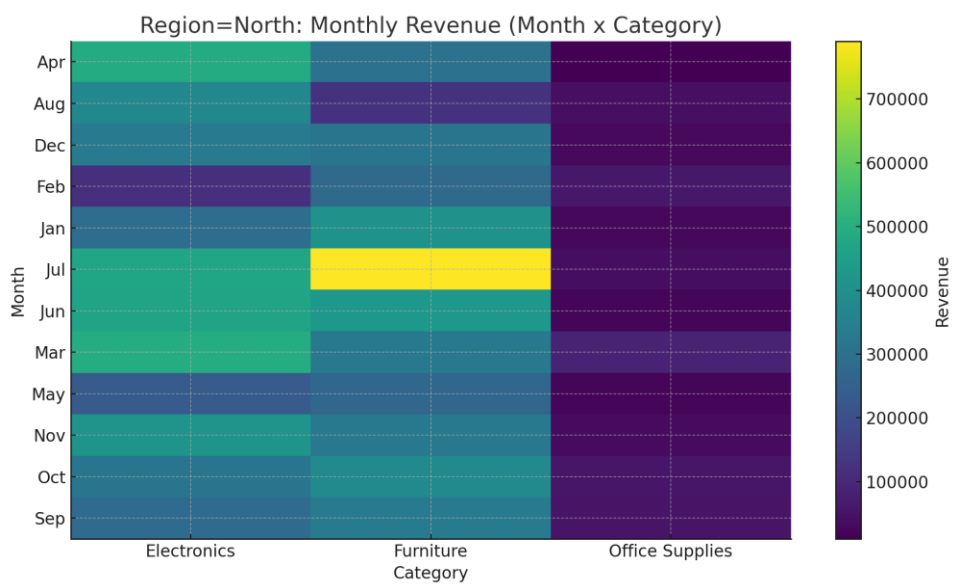
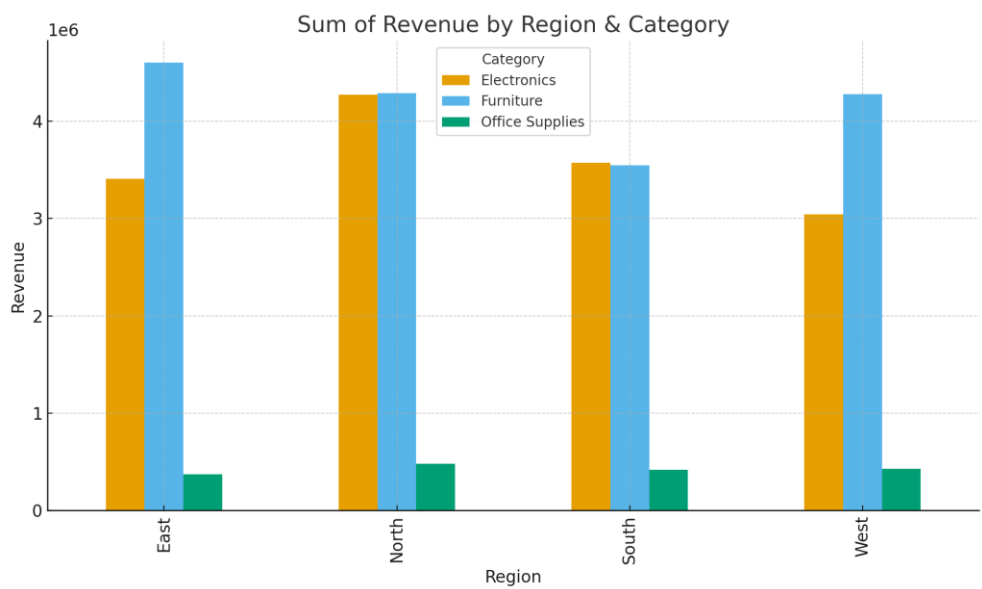
plt.xlabel("Category"); plt.ylabel("Month")

plt.colorbar(label="Revenue")

plt.show()
```

Date	Region	Store	Category	Units	UnitPrice	Revenue
31-01-2025	North	S04	Electronics	63	1280.99	80702.65
31-01-2025	North	S04	Furniture	9	4393.06	39537.51
31-01-2025	North	S04	Office Supplies	27	23.05	622.45
31-01-2025	North	S05	Electronics	42	4025.93	169089.2
31-01-2025	North	S05	Furniture	66	2446.09	161441.8
31-01-2025	North	S05	Office Supplies	66	181.49	11978.13
31-01-2025	North	S06	Electronics	27	1423.37	38431.1
31-01-2025	North	S06	Furniture	79	2621.83	207124.7
31-01-2025	North	S06	Office Supplies	38	341.03	12959.08
31-01-2025	South	S03	Electronics	51	4947.01	252297.4
31-01-2025	South	S03	Furniture	39	969.02	37791.69
31-01-2025	South	S03	Office Supplies	21	375.27	7880.73

31-01-2025	South	S05	Electronics	13	371.27	4826.45
31-01-2025	South	S05	Furniture	8	2671.47	21371.73
31-01-2025	South	S05	Office Supplies	77	551.96	42500.71
31-01-2025	South	S06	Electronics	39	3220.29	125591.2
31-01-2025	South	S06	Furniture	38	2584.99	98229.71
31-01-2025	South	S06	Office Supplies	43	163.56	7033.02
31-01-2025	East	S06	Electronics	56	1162.91	65123.09
31-01-2025	East	S06	Furniture	59	217.92	12857.54



PRACTICAL NO 6

6

CONNECTING AND EXTRACTING WITH VARIOUS DATA RESOURCES IN TABLEAU AND PERFORM CALCULATIONS AND CREATING PARAMETERS IN TABLEAU

Solution:

Connecting & Extracting Data, Calculations, and Parameters (using *Sample – Superstore*)

Task:

Using the Excel file “**Sample – Superstore.xls**”:

1. Connect to the dataset and set up relationships between tables.
2. Create an **extract** for performance.
3. Build **three calculated fields**:
 1. Profit Ratio
 2. Region – Category (string concat)
 3. Year of Order
4. Create a **parameter** “Profit Threshold” and use it in a calculation to classify results as **High/Low**.
5. Create two visualizations and a **dashboard** that demonstrates the effect of the parameter.

Prerequisite:

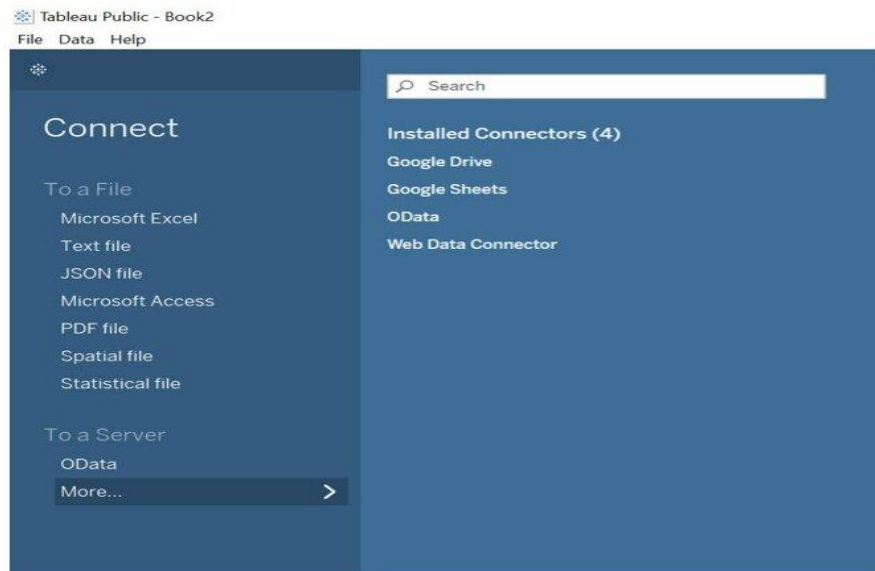
Download the dataset “**Sample – Superstore.xls**” from Tableau’s official website:

<https://www.tableau.com/sites/default/files/2021-05/Sample%20-%20Superstore.xls>

Step 1: Connect & Relate Tables

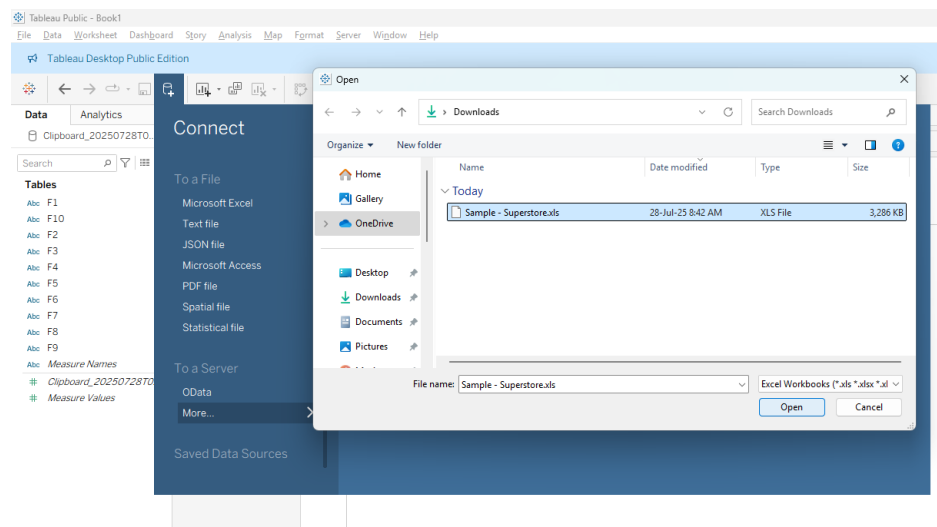
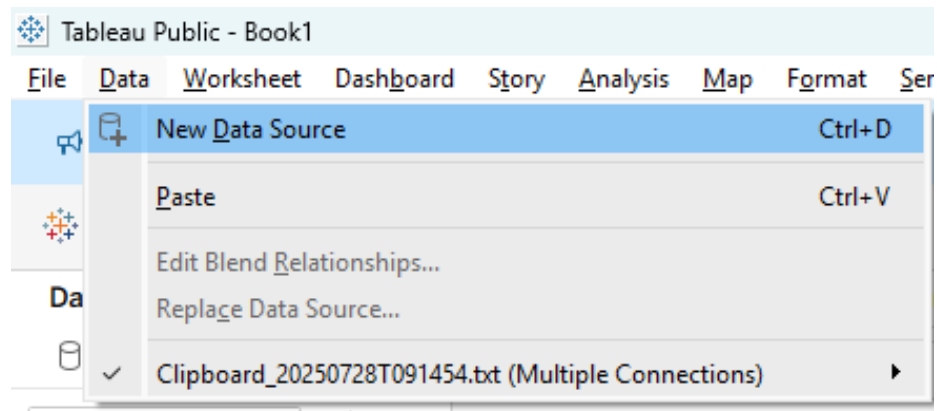
As soon as you launch the tableau you will get:

Data Visualization Practical

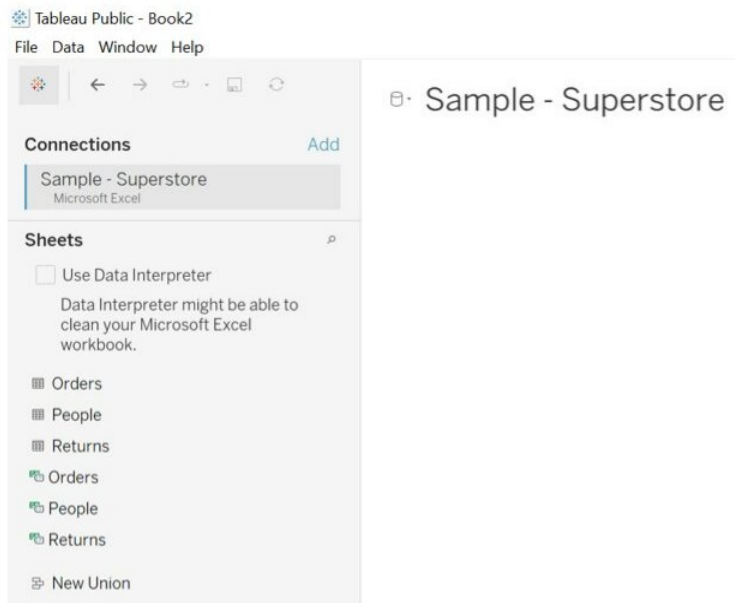


Connect to Sample - Superstore Data

Navigate to the **Data** menu choose **New Data Source** to open the Connect pane. Click on **Microsoft Excel**, then locate and select the saved Excel file (**Sample - Superstore.xlsx**) on your computer to open it.

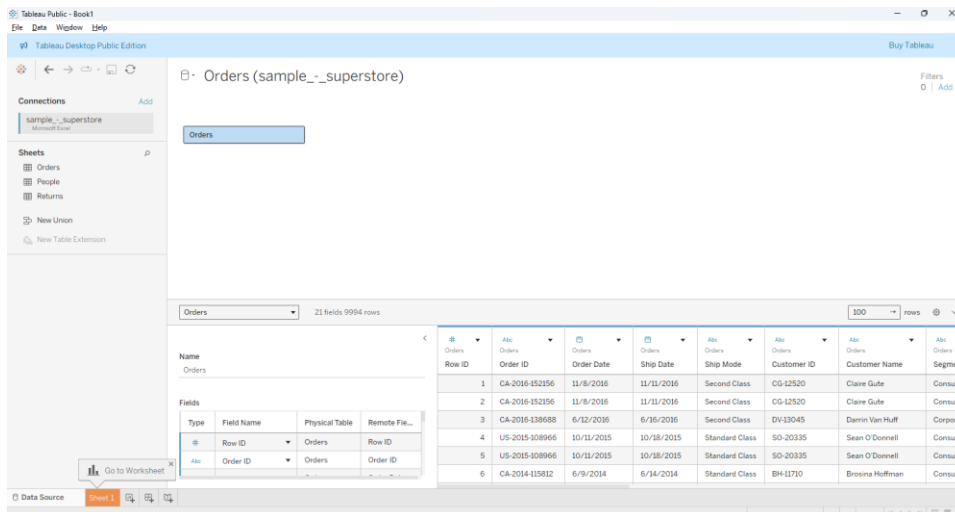


It connects your excel file in Tableau.



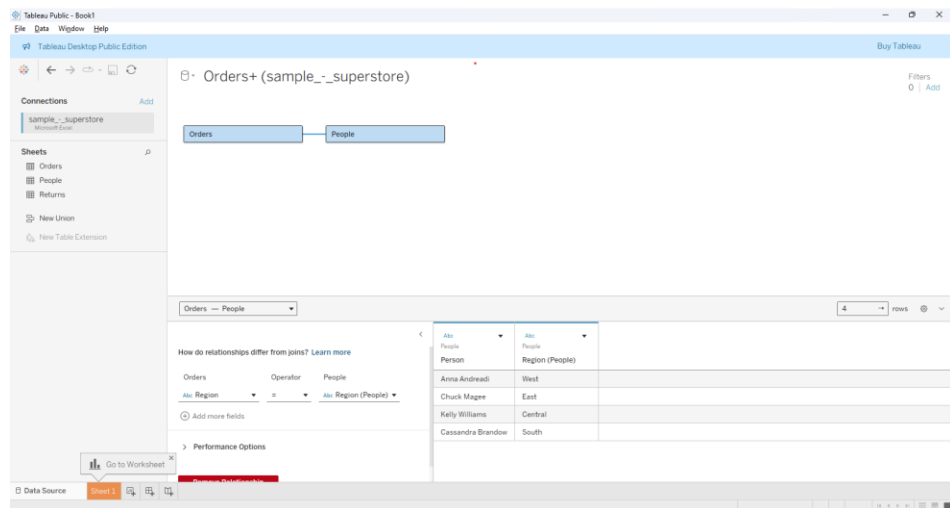
Connecting and
Extracting with
Various data
Resources in Tableau
and Perform
Calculations and
Creating Parameters
in Tableau

Drag **Orders** to the canvas (main table).

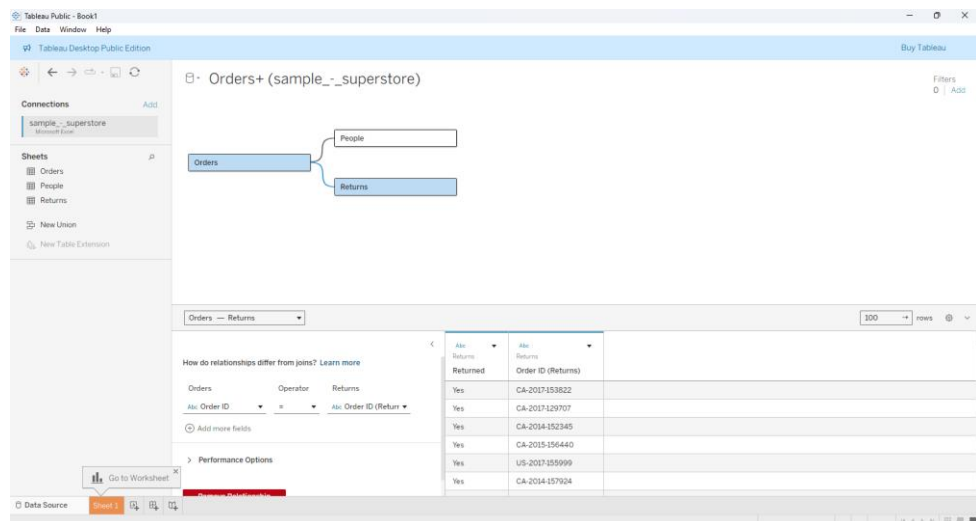


Drag **People** next to Orders → **Relationship** on **Region = Region** (leave Cardinality/Performance as default).

Data Visualization Practical

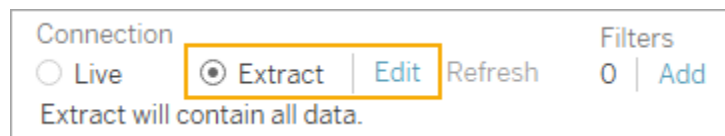


Drag **Returns** and relate **Returns[Order ID] = Orders[Order ID]**



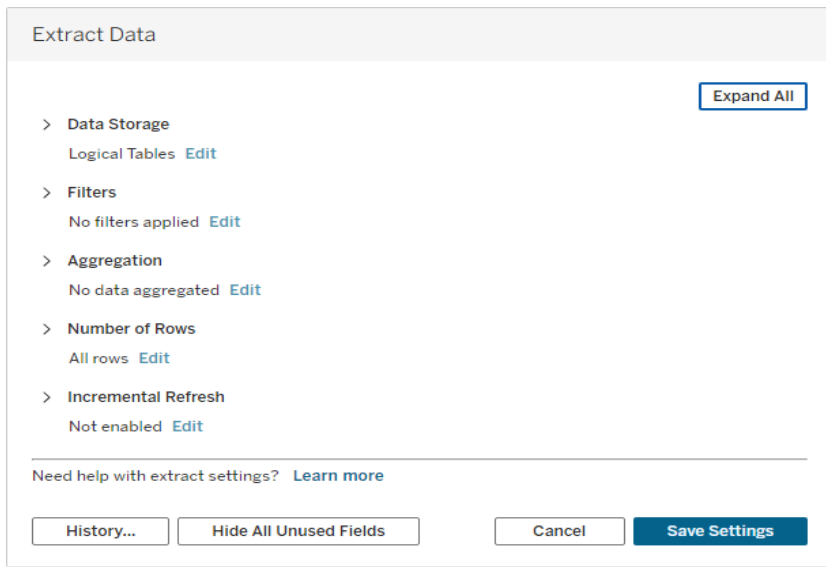
Step 2: Create an Extract

Once you've connected to your data and configured the data source on the **Data Source** page, go to the top-right corner, choose **Extract**, and then click the **Edit** link to open the **Extract Data** dialog box.



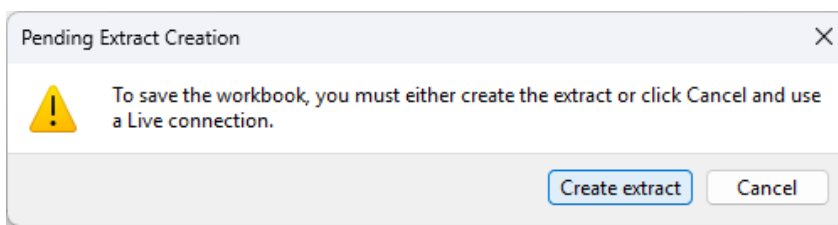
After choosing **Extract**, go to any worksheet (Sheet 1).

Connecting and
Extracting with
Various data
Resources in Tableau
and Perform
Calculations and
Creating Parameters
in Tableau

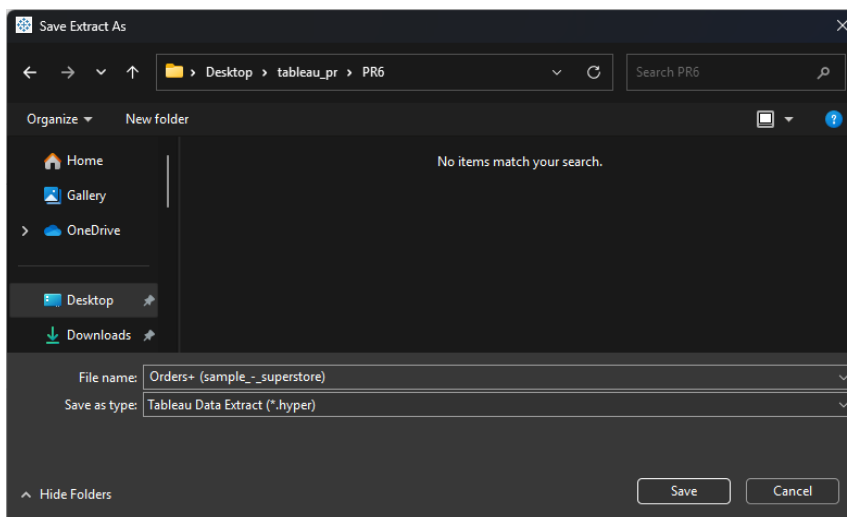


Click on Save Settings

Goto Save Option and select Create extract

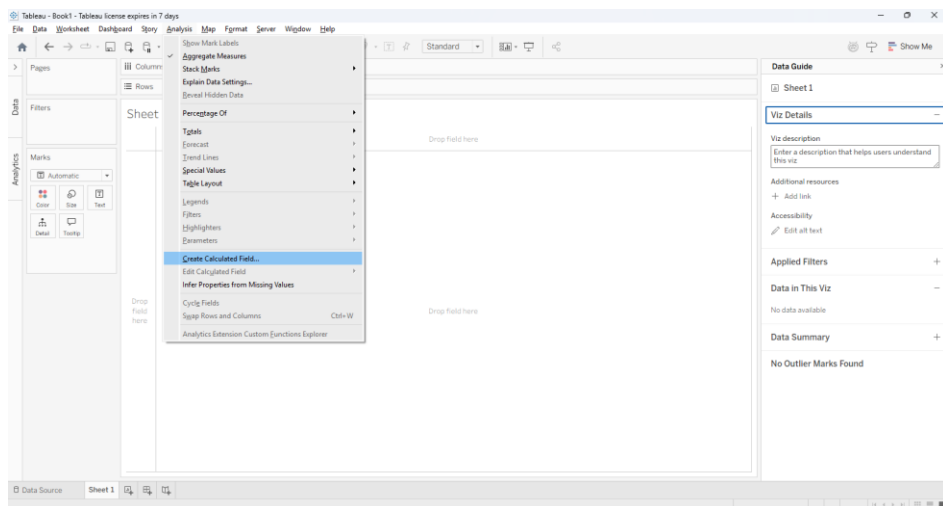


Select the desired location, Give name to the file and click on save.
It will create a .hyper file.

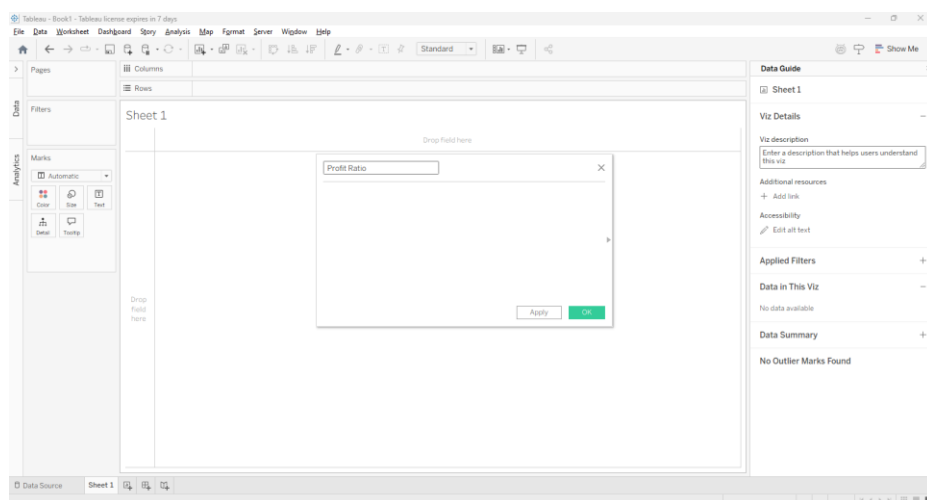


Step 3: Create CalculatedFields

In Tableau, select Analysis > Create Calculated Field.



Enter a name for the calculated field. In this example, the field is called, Profit Ratio



Enter a formula. This example uses the following formula:



Connecting and
Extracting with
Various data
Resources in Tableau
and Perform
Calculations and
Creating Parameters
in Tableau

Click ok

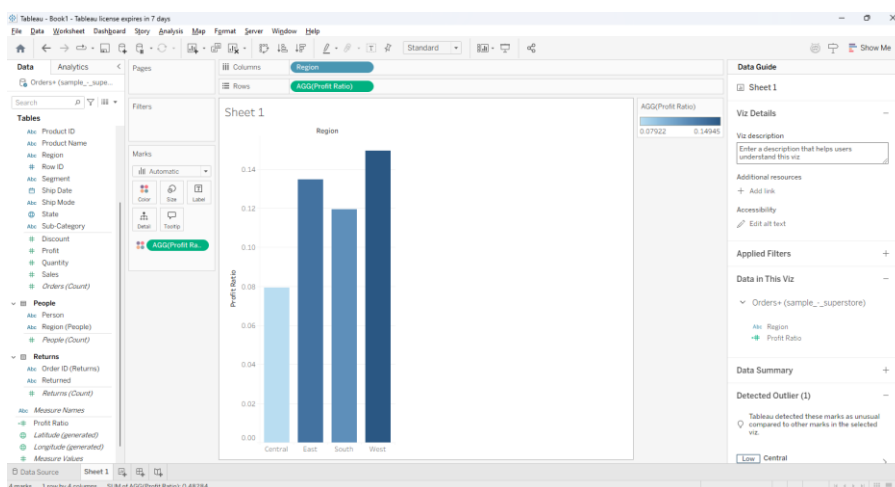
When finished, click OK. The new calculated field is added to Data pane as a measure because it returns a number.

This shows how profitable the sales are as a percentage of total sales. A high profit ratio means sales are generating good profit; a low ratio means costs are high or margins are small.

To Build Views with Parameter

1. Profit Ratio by Region (with Parameter Classification)

1. Drag **Region** to Columns.
2. Drag **Profit Ratio** to Rows.
3. Drag **Profit Status** to Color.



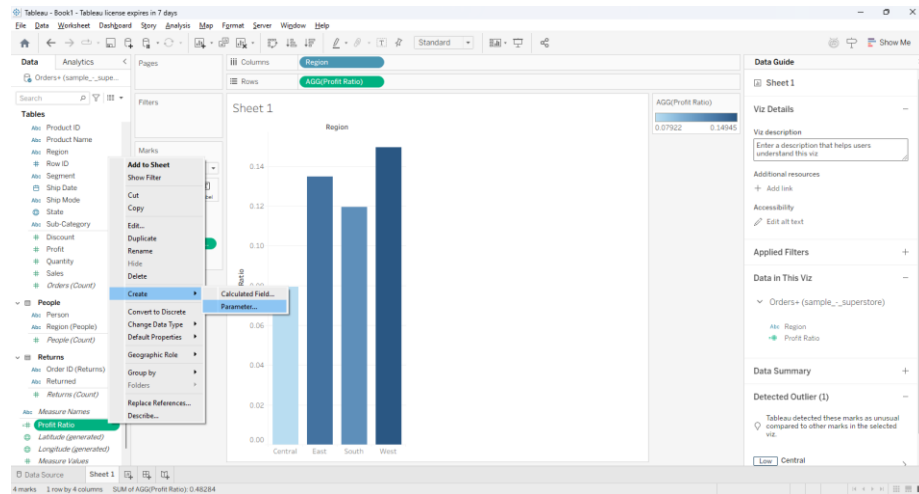
A bar chart showing Profit Ratio by Region.

Step 4 : Create a **parameter** “Profit Threshold” and use it in a calculation to classify results as **High/Low**.

Open Parameter Dialog Box

1. In the Data Pane (left side of Tableau interface, where dimensions and measures are listed):

1. Right-click on Profit Ratio → Create Parameter.



Define the Parameter Properties
In the Create Parameter dialog box:

Name: Profit Threshold

1. Data Type: Choose Float (Decimal Number)
(Since profit ratios are in decimals, e.g., 0.10 = 10%)

2. Current Value: Enter 0.10 (this means the default threshold is 10%)

1. Allowable Values: All → User can type any value.

2. List → Provide specific values (e.g., 0.05, 0.10, 0.20).

3. Range → Set min, max, and step size. Example:

1. Minimum: 0.00

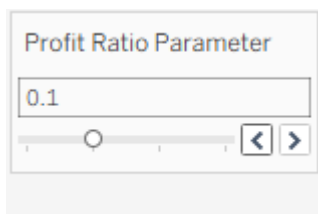
2. Maximum: 0.50

3. Step size: 0.05

For this practical, choose Range with step size 0.05.

Add the Parameter to the View

1. Once created, the parameter will appear in the Parameters section (bottom left of the Data Pane).
2. Right-click the parameter → Show Parameter Control.
3. The control will now appear on the worksheet (usually on the right side as a slider or dropdown).



Customize the Control

By default, Tableau shows a dropdown.

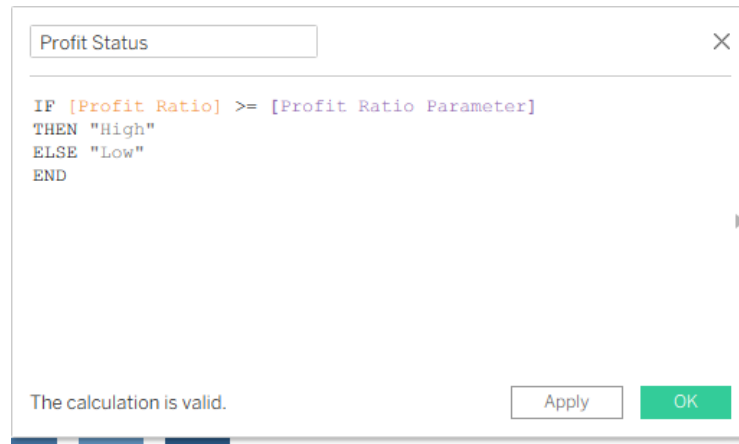
To make it interactive:

1. Click the dropdown arrow on the control → choose **Slider**.
2. Now you can adjust the threshold dynamically by sliding between values.
3. This parameter lets us define a **custom profit threshold**.

It is later used in a **calculated field** to classify results as **High** or **Low profit**.

Example calculation:

Create a calculation Field with name **Profit Status**



Click OK.

Now, Profit Status will appear as a new field in your Data Pane (under Dimensions).

Use the Field in a View

Drag **Region** → Columns.

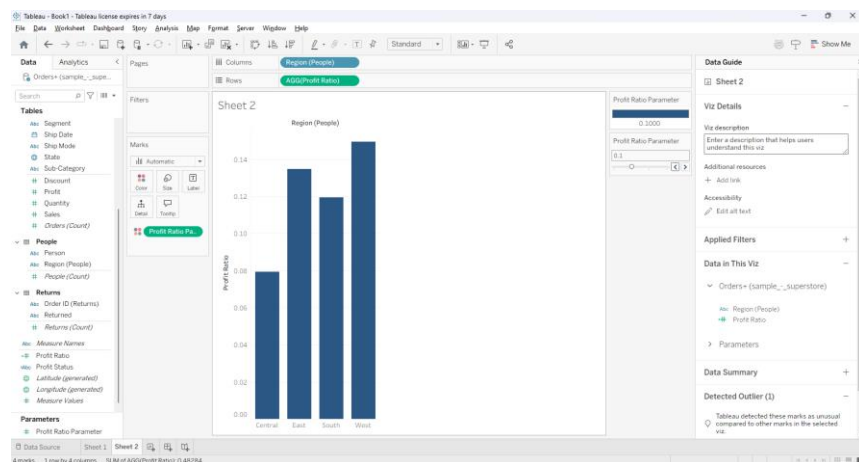
Drag **Profit Ratio** → Rows.

Drag the new **Profit Status** field → Color (on Marks card).

Show the Parameter Control

In the **Data Pane**, right-click on **Profit Threshold** → **Show Parameter Control**.

A slider/dropdown will appear on the worksheet (usually top-right).



Move the slider (or change the dropdown value).

Tableau will re-evaluate the calculation:

If Profit Ratio $\geq$ chosen threshold → Bar is colored as **High**.

Otherwise → Bar is **Low**.



PRACTICAL 7

7

DESIGNING TABLEAU DASHBOARDS FOR DIFFERENT DISPLAYS AND DEVICES

Solution:

A Dashboard in Tableau is a collection of views (charts, maps, tables) displayed together for comparative analysis.

Step 1. Create Worksheets

1. Connect to Sample – Superstore dataset.

2. Create at least two visualizations:

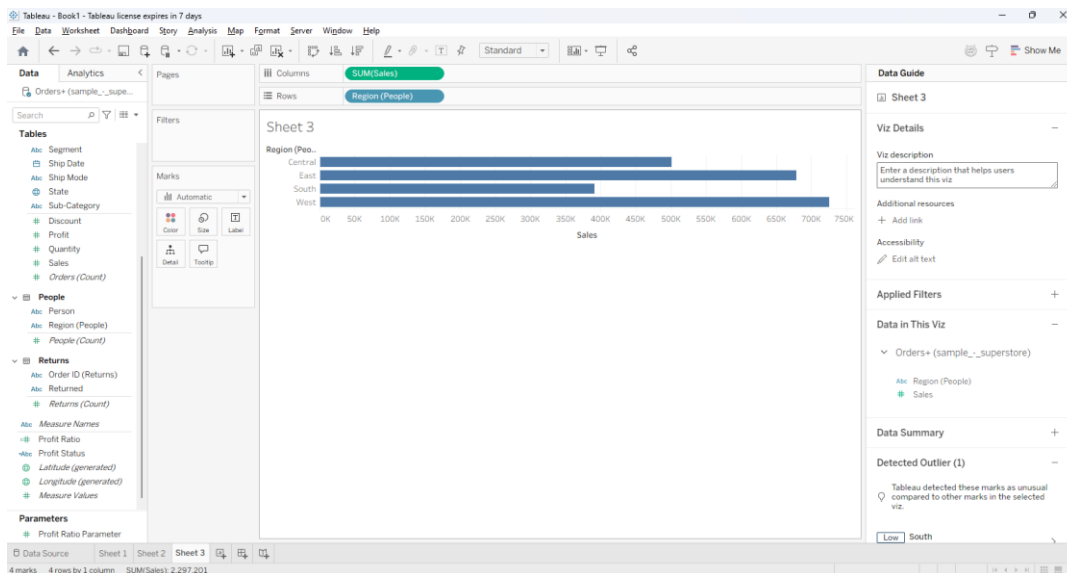
3. **Sales by Region (Bar Chart)**

Go to a **new worksheet** (click on the Worksheet tab at the bottom).

From the **Data Pane** (left side):

Drag **Region** → to **Rows**. and Drag **Sales** → to **Columns**.

Tableau will automatically create a **horizontal bar chart** showing total Sales for each Region.



4. Profit by Year (Line Chart)

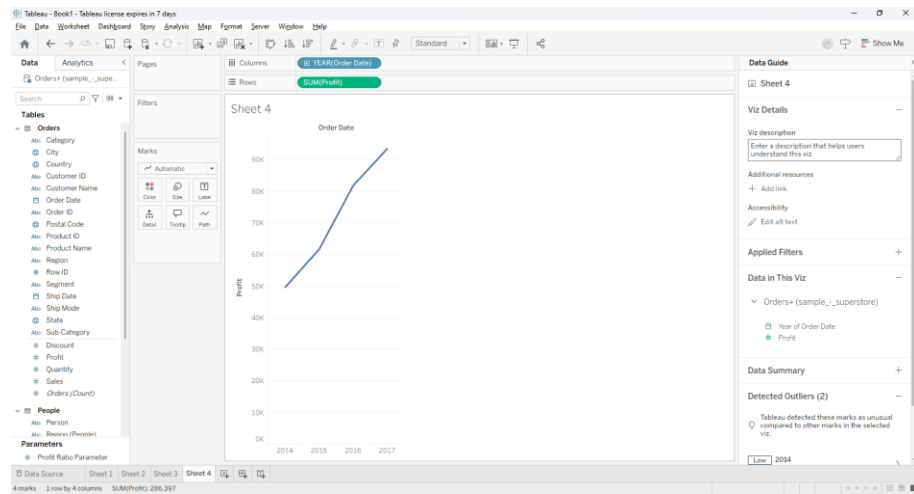
Open a **new worksheet**.

From the **Data Pane**:

Drag **Order Date** → to **Columns**. Tableau will show years by default.

Drag **Profit** → to **Rows**.

Tableau creates a **line chart** showing Profit trend over the years.



5. Save them as separate worksheets.

Step 2. Create a Dashboard

6. Click **New Dashboard** (bottom tab).

7. From the left pane, drag the created worksheets into the dashboard canvas.

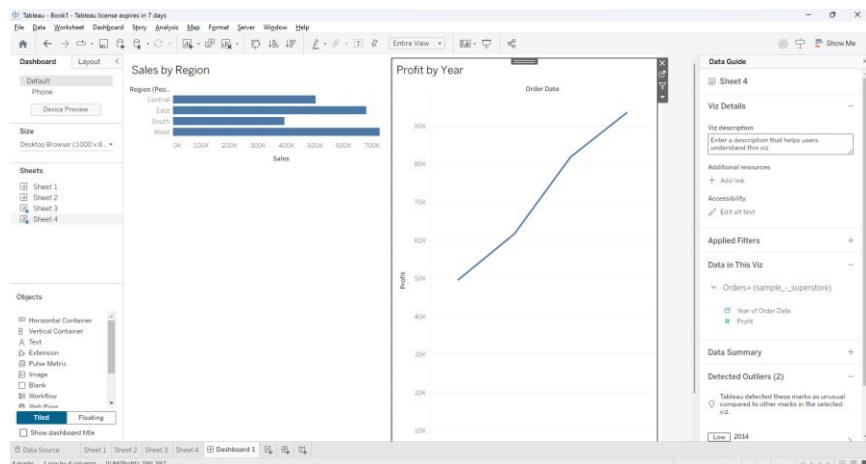
8. Arrange them side by side or top-bottom layout.

9. Add interactive elements:

1. **Filters** (e.g., Category filter)

2. **Parameter controls**

3. **Legends**



Step 3. Use Device Designer

10. On the top menu → **Dashboard** → **Device layouts**.
11. Choose **Device type**: Desktop, Tablet, Phone.

Designing Tableau
Dashboards for
different displays and
devices

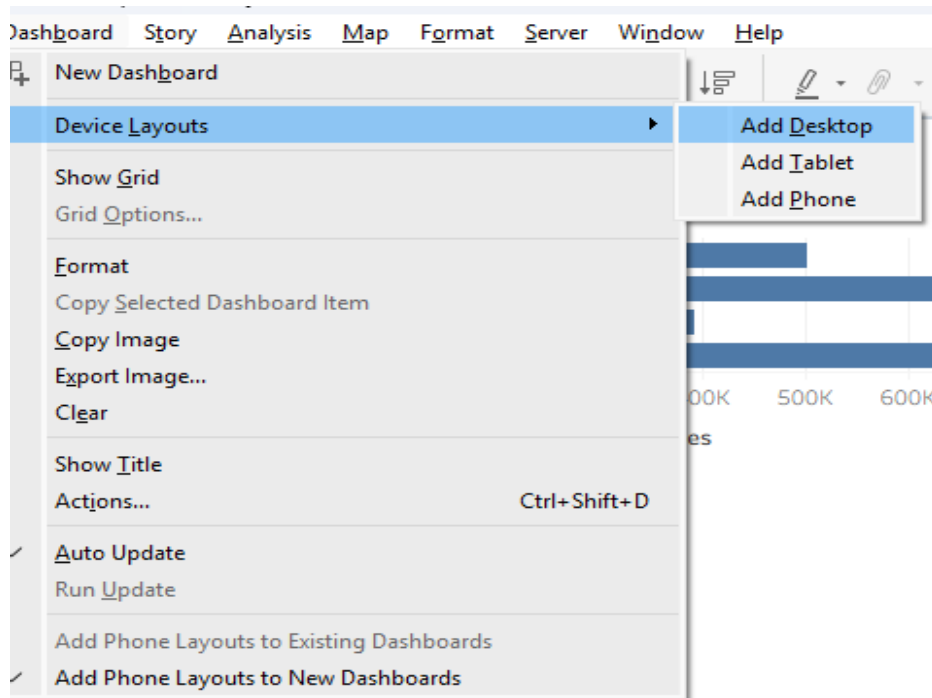
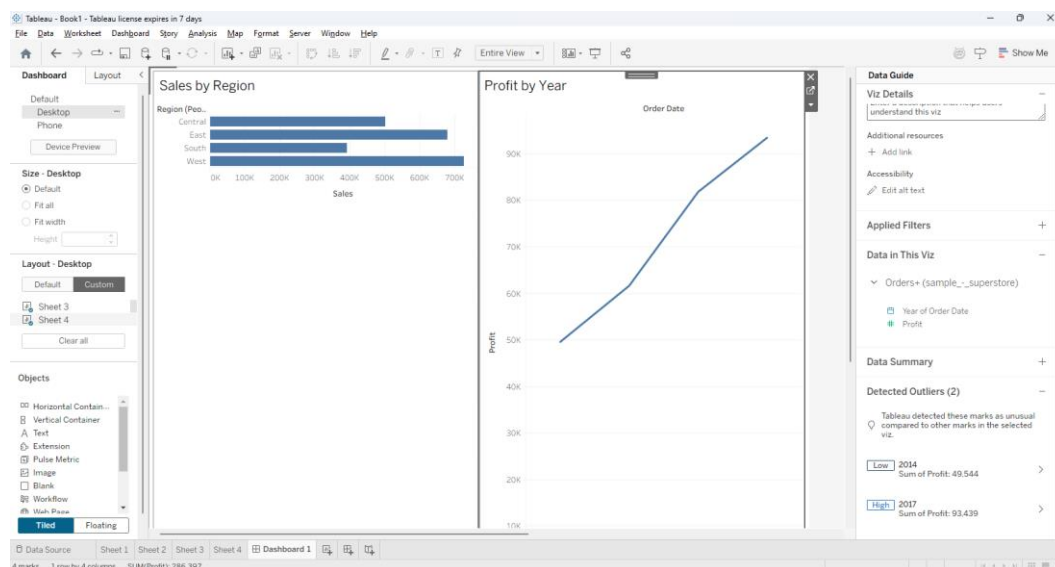
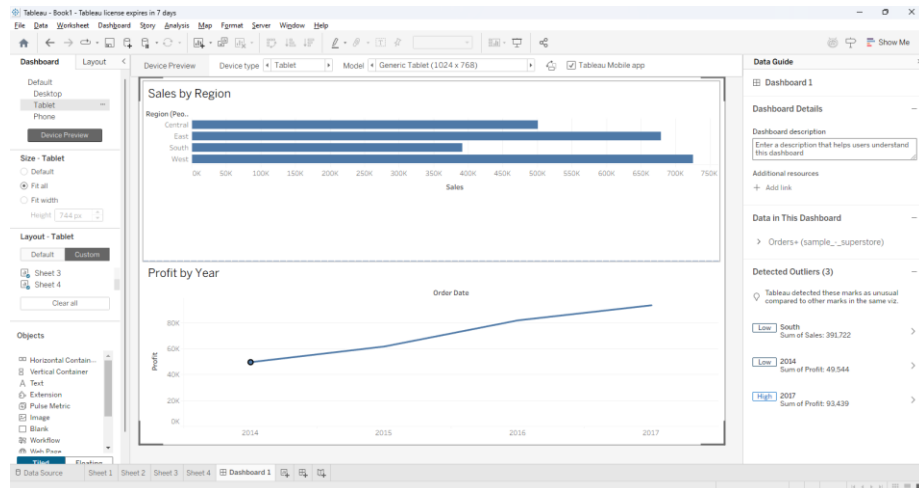


Tableau will display a preview frame for that device.

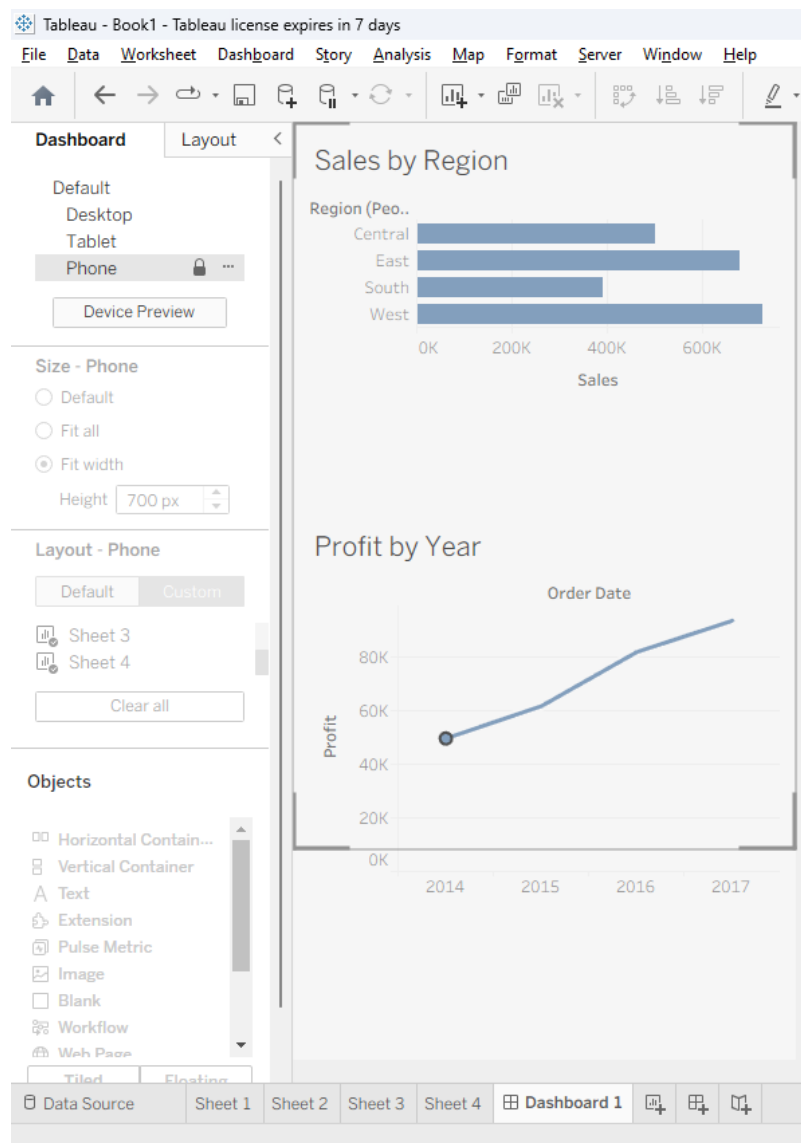
For Desktop: Keep all charts, filters, and legends visible.



For Tablet: Adjust layout (stack charts vertically for better fit).



For Phone: Show only the most important chart (use scrolling design).



PRACTICAL 8

8

CREATE A TREND MODEL USING DATA, ANALYSE-IT AND USE IT FOR FORECASTING

Solution:

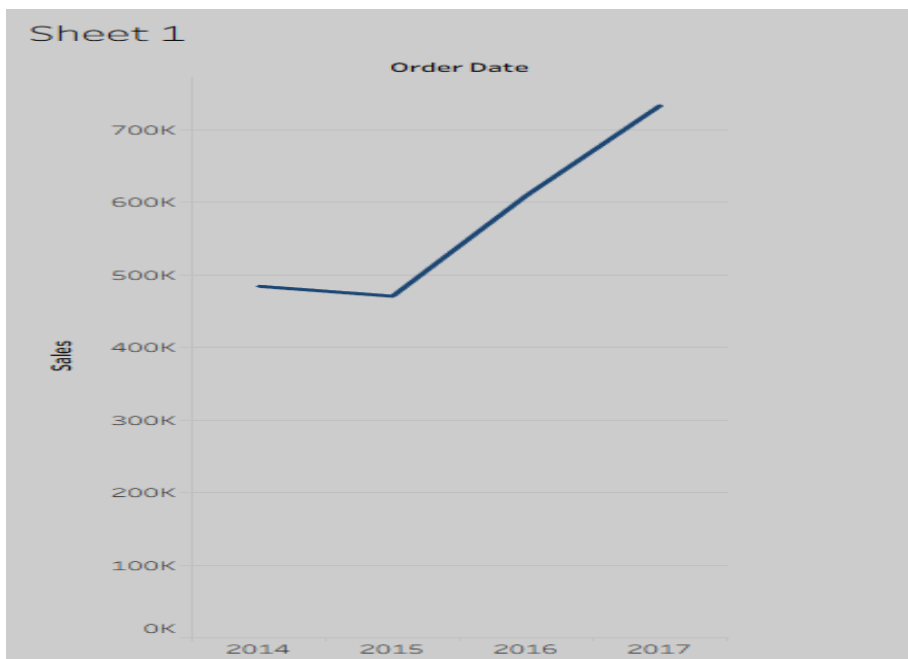
To build a trend model using Tableau and apply it for forecasting future values.

Step 1. Connect Data

1. Open **Tableau Desktop 25.2**.
2. Connect to the dataset (e.g., **Sample – Superstore.xls**).
3. Go to the worksheet.

Step 2. Create a Time Series View

4. Drag **Order Date** to **Columns** (right-click → select *Year* or *Month*).
5. Drag **Sales** (or Profit) to **Rows**.
You now have a **line chart** showing sales over time.

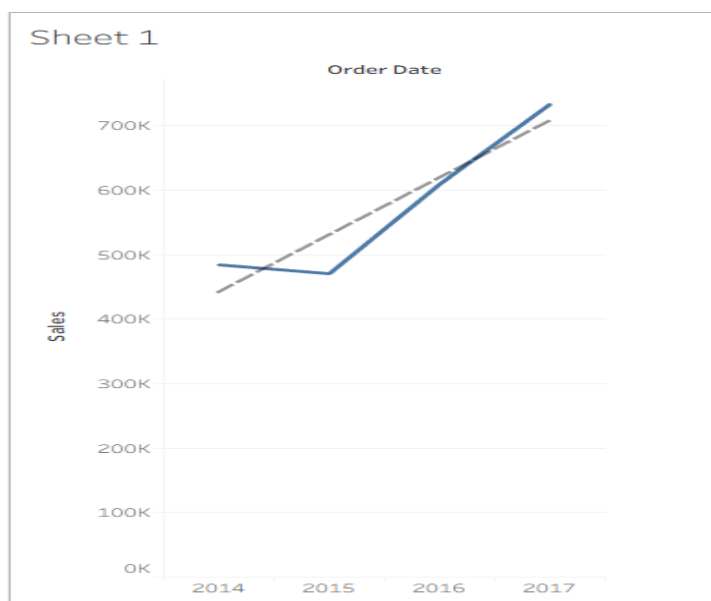


Step 3. Add a Trend Line

- On the top menu, go to **Analysis** → **Trend Lines** → **Show Trend Lines**.

(or open the **Analytics** pane → drag **Trend Line** into view).

- By default, Tableau applies a **linear trend line**.



- To change or customize:

- Go to **Analysis** → **Trend Lines** → **Edit Trend Lines...**

- A dialog box will open where you can select:

The 'Trend Lines Options' dialog box is shown. It has a title bar with a close button (X). The 'Model Type' section contains radio buttons for 'Linear' (selected), 'Logarithmic', 'Exponential', 'Power', and 'Polynomial'. The 'Polynomial' option has a 'Degree' dropdown set to '2'. The 'Factors' section has a text box with the label 'Build separate trend lines based on the following dimensions:'. The 'Options' section contains checkboxes for 'Show tooltips' (checked), 'Show confidence bands' (unchecked), 'Allow a trend line per color' (checked), 'Show recalculated line for highlighted or selected data points' (checked), and 'Force y-intercept to zero' (unchecked). An 'OK' button is at the bottom right.

1. **Model Type: Automatic** → Tableau chooses the best model type based on the data pattern.

Create a Trend model using data, Analyse-it and use it for forecasting

Linear → Fits a straight line showing constant rate of increase or decrease.

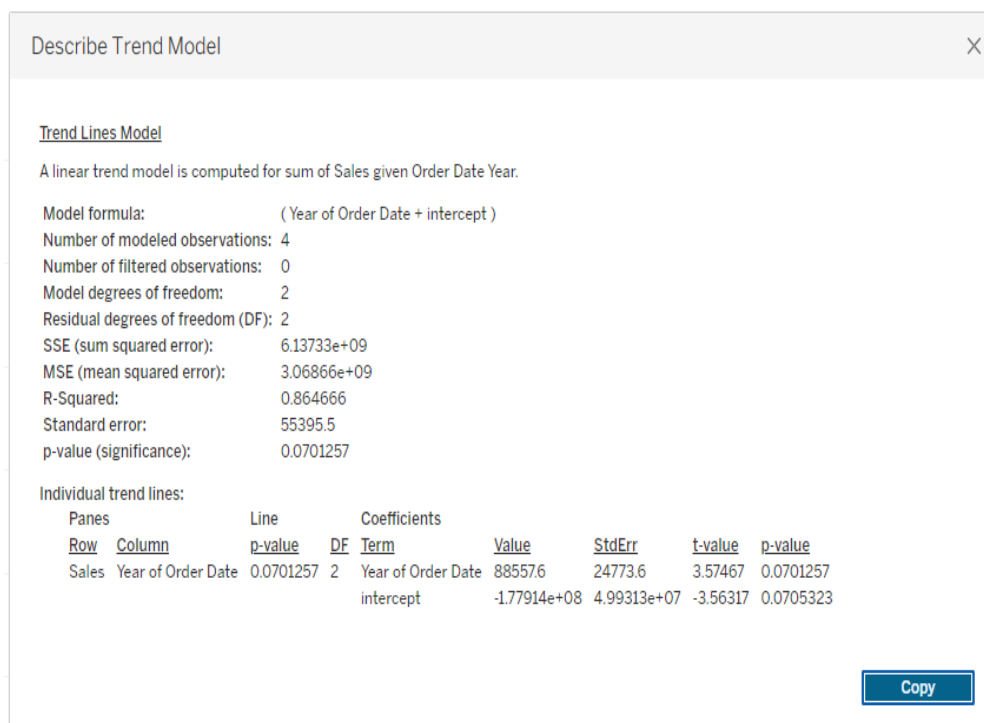
Logarithmic → Models rapid change at the beginning that slows over time.

Exponential → Models constant percentage growth or decay (curved upward/downward).

Polynomial → Fits a curved line that can capture more complex fluctuations.

9. To view details:

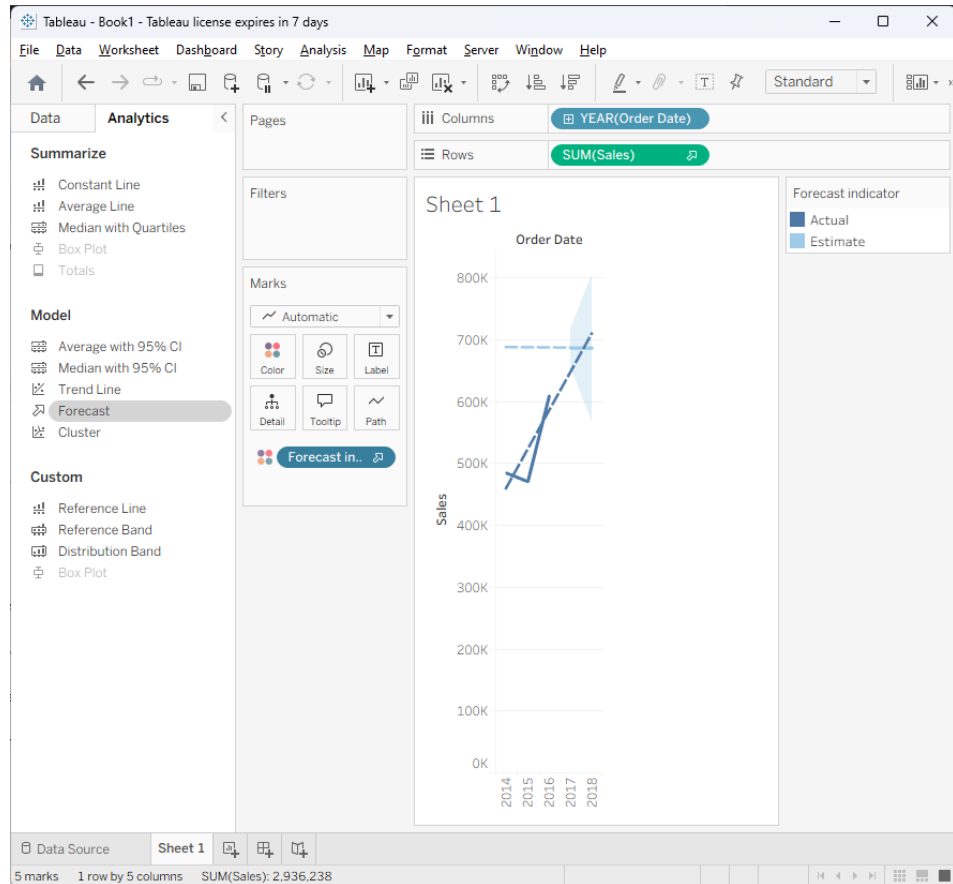
1. Go to **Analysis** → **Trend Lines** → **Describe Trend Model**.
2. This gives equation, R^2 value, and statistical significance.



Step 4. Add Forecasting

Go to the **Analytics pane** (left sidebar). Drag **Forecast** onto your view. Tableau will automatically generate a forecast line based on the historical trend.

Data Visualization Practical



To customize:

Right-click on the forecast line → **Forecast** → **Forecast Options**.
In the dialog box, you can set:

Forecast Length → e.g., 12 months, 8 quarters, or 2 years ahead.

Model Type → Automatic, Linear, or Exponential.

Seasonality → Automatic, None, or Custom (for monthly/quarterly/seasonal data).

Forecast Line (dashed) → Tableau's predicted values for future periods.

Confidence Bands (shaded area) → range of uncertainty, showing high/low possible outcomes.

If you right-click → **Forecast** → **Describe Forecast**, Tableau explains the model used, seasonal effects, and accuracy.



PRACTICAL 9

9

CREATING GEOSPATIAL FEATURE MAPS IN TABLEAU USING GEOSPATIAL DATA

Solution:

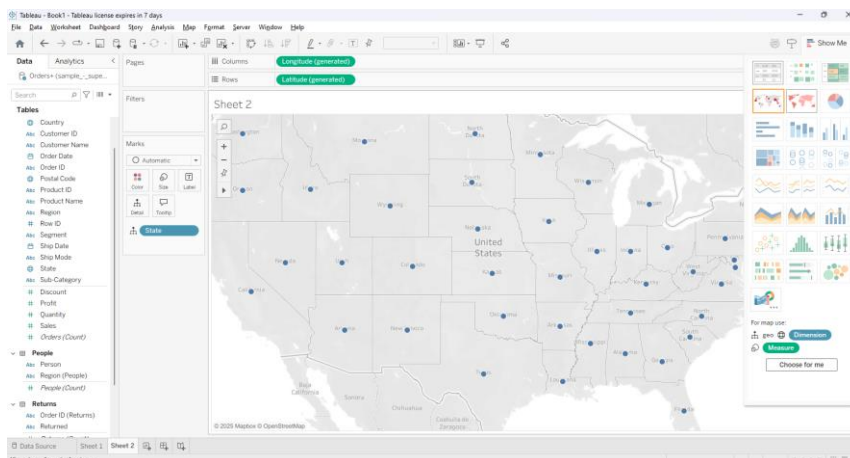
To visualize and analyze geographic (spatial) data in Tableau by creating feature maps.

Step 1. Connect to Data

1. Connect to the dataset (**Sample – Superstore.xlsx**).
2. Go to the **Data Source** tab and ensure geographic fields (**Country**, **State**, **City**) are recognized with a **globe icon** 🌐.
3. If not, right-click field → **Geographic Role** → **Assign** (e.g., Country, State, City, Postal Code).

Step 2. Create a Basic Map

1. Drag **State** (or City) to **Rows**.
2. Tableau automatically generates a **map view**.
3. If not: From the top menu → **Show Me** → select **Map**.



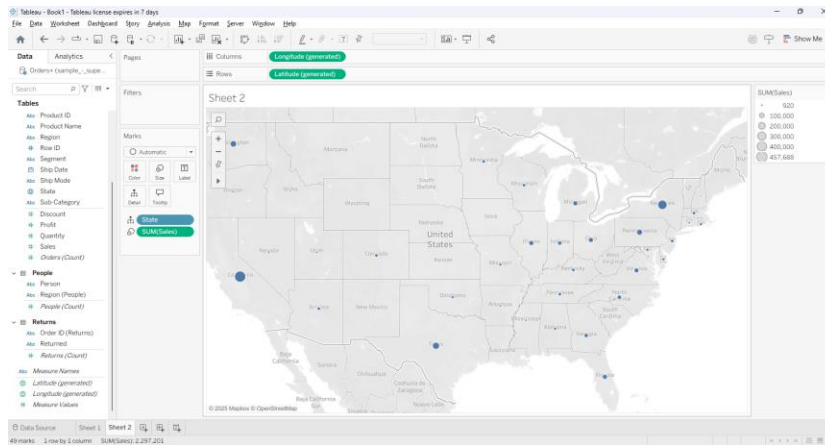
Filled map showing Sales by State.

Step 3. Add Measure to Map

1. Drag **Sales** (or Profit) to **Size** on Marks card.

2. Drag **Profit** to **Colour**.

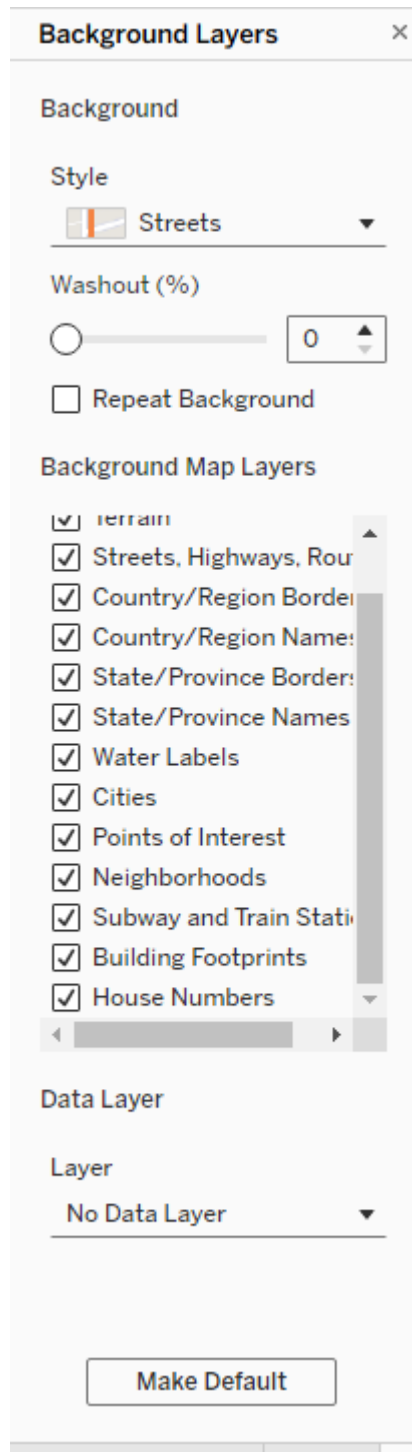
1. Now the map shows circles sized by Sales and coloured by Profit.



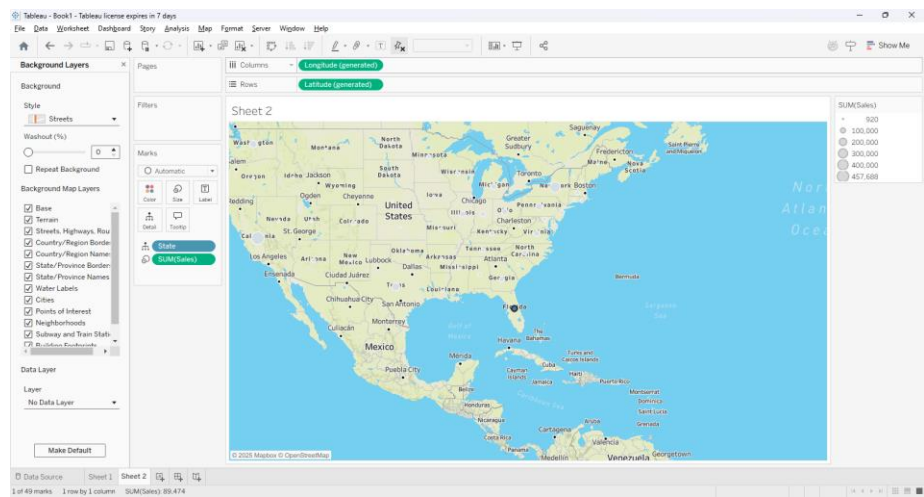
Bubble map showing Profit by City.

Step 4. Customize Geographic Layers (Tableau 25.2)

1. On the **top menu**, go to:
Map → Background Layers.
2. You will see options to turn ON/OFF different map features such as:
 1. **Streets and Highways**
 2. **Country Borders**
 3. **Zip Code Boundaries**
4. You can also change the **Background Map Style**:
 1. **Normal** (default)
 2. **Dark**
 3. **Satellite**
5. Adjust layers and style based on your visualization needs.
 1. Example: Use **Dark background** for highlighting colorful data points.
 2. Use **Satellite style** for real-world geography look.



Data Visualization
Practical



Interactive dashboard with geographic insights.



CREATE DASHBOARD AND STORYTELLING IN TABLEAU

Solution:

To design a dashboard combining multiple visualizations and use storytelling features to present insights effectively.

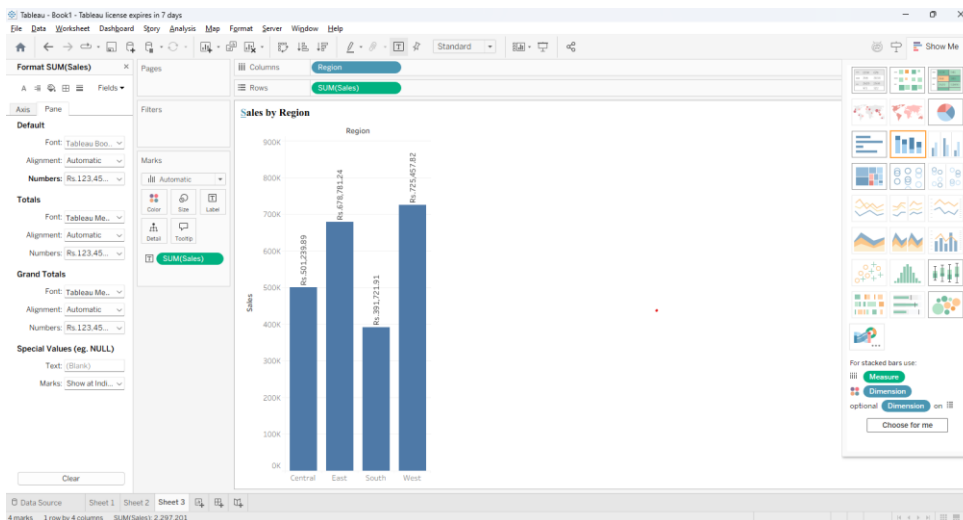
Prepare Visualizations

Before creating a dashboard, you must first create individual **worksheets (charts)**. Let's use the Sample – Superstore dataset as an example.

Step 1. Visualization 1: Sales by Region (Bar Chart)

1. Open a new worksheet.
2. Drag Region → Columns.
3. Drag Sales → Rows.
4. Tableau will generate a vertical bar chart.
5. From the top menu → Show Me → choose Bar Chart (if not already selected).
6. Add Labels: Drag Sales again onto Label.
7. Format: Right-click axis → Format → Change to currency for better readability.

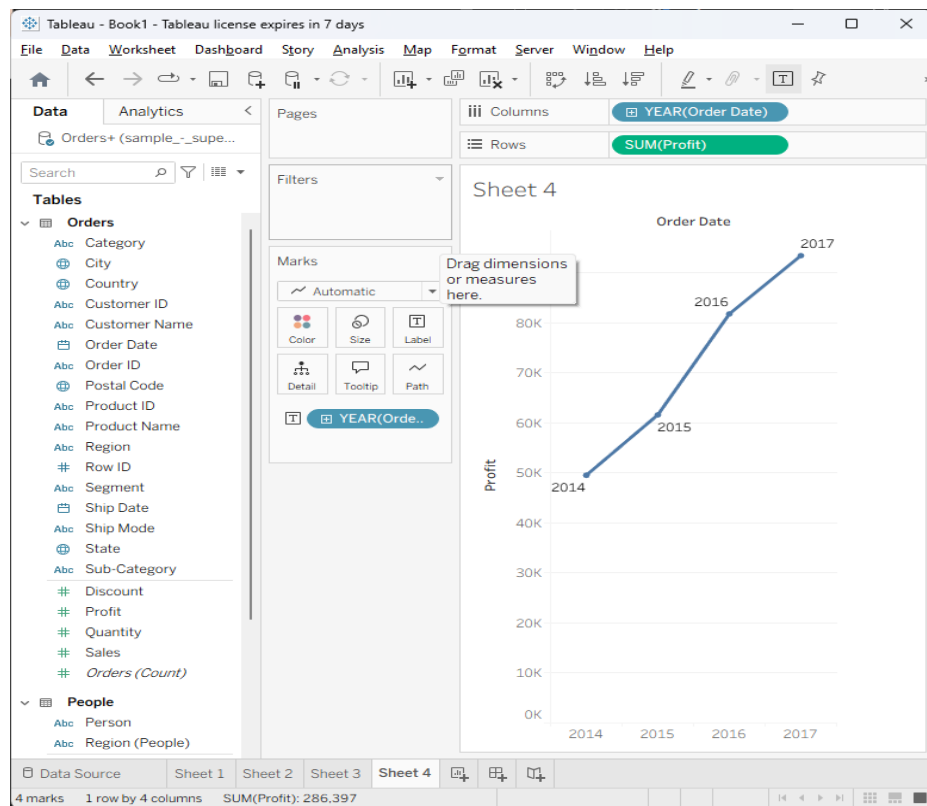
This chart shows total sales per Region.



Step 2. Visualization 2: Profit Trend by Year (Line Chart)

1. Open another new worksheet.
2. Drag Order Date → Columns.
3. Right-click → select Year(Order Date).
4. Drag Profit → Rows.
5. Tableau will generate a line chart showing profit trend.
6. Add Labels/Tooltips for clarity.
7. Format the X-axis → ensure years display clearly.

This chart shows profit changes over time.

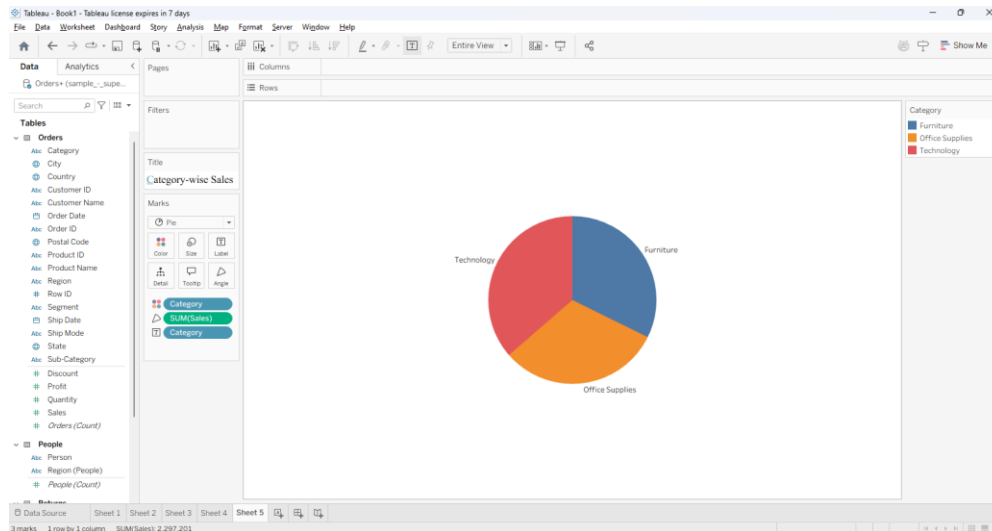


Step 3. Visualization 3: Category-wise Sales (Pie Chart) (Optional, for variety)

1. Open a new worksheet.
2. Drag Category → Colour.
3. Drag Sales → Angle.

4. From the Show Me panel → select Pie Chart.
5. Drag Category → Label to display names.
6. Adjust size with the Size shelf.

This chart shows sales share of each product category.



Now you have at least 2–3 visualizations ready, which can be combined into a dashboard in the next step.

Renaming Worksheets

1. At the bottom of Tableau, each sheet is shown as a **tab** (like in Excel).
Example: Sheet 1, Sheet 2, etc.
2. **Double-click** on the sheet tab name.
3. Type a meaningful name, e.g.:
 1. Sales by Region
 2. Profit Trend
 3. Forecast Analysis
 4. Customer Map

Step 4. Create a Dashboard

On the menu bar → **Dashboard** → **New Dashboard**.

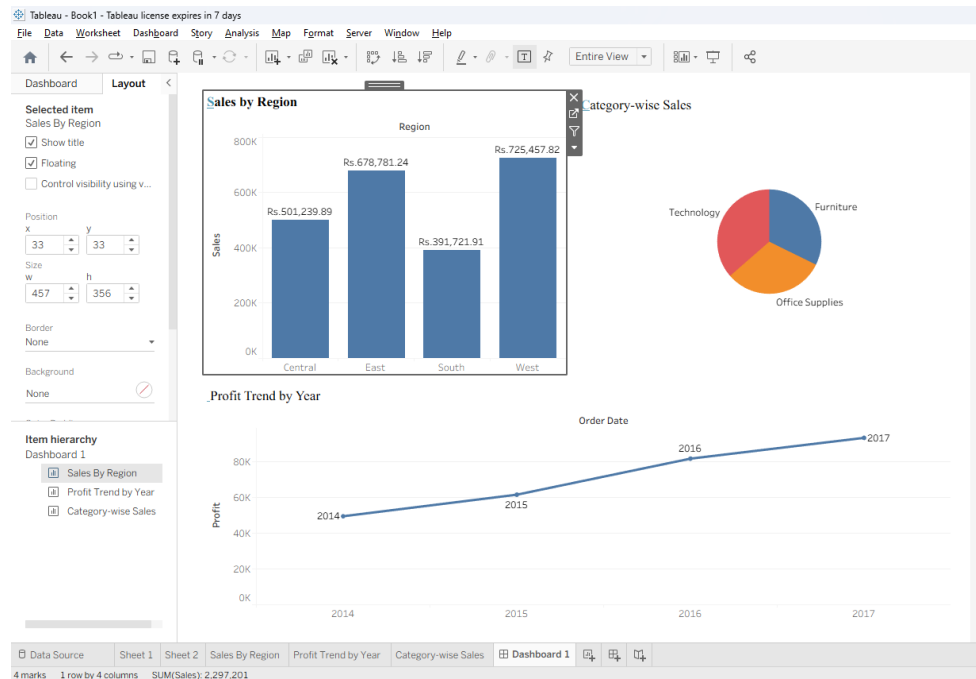
A blank dashboard workspace opens.

From the left **Sheets pane**, drag your visualizations into the dashboard.

Arrange them using **Tiled** or **Floating** layout.

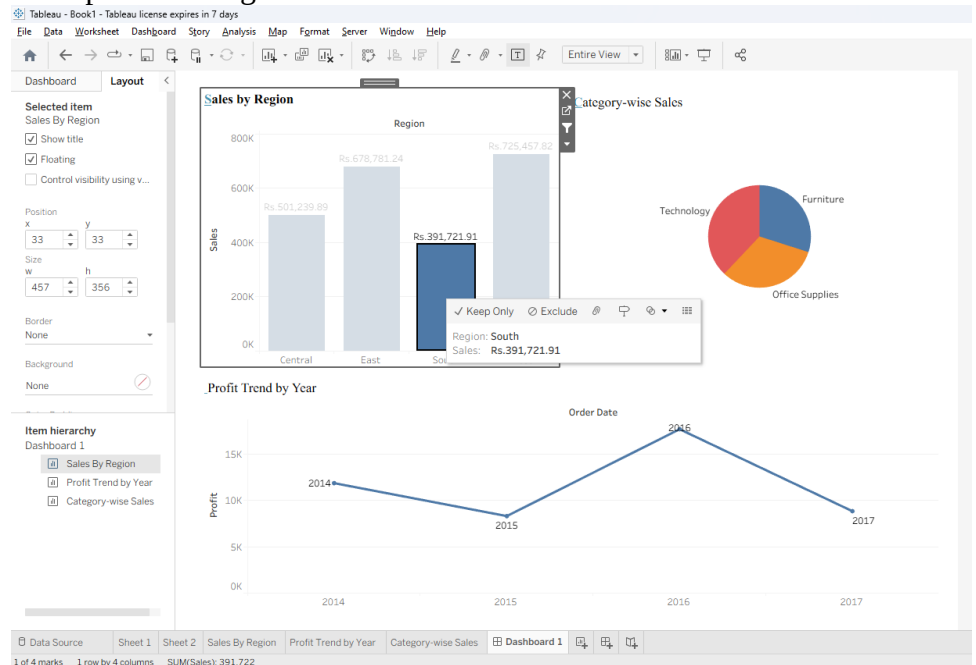
Data Visualization Practical

4. Tiled → Fits charts in a grid-like structure.
5. Floating → Allows free positioning like layers.



Add **Filters and Legends** if needed:

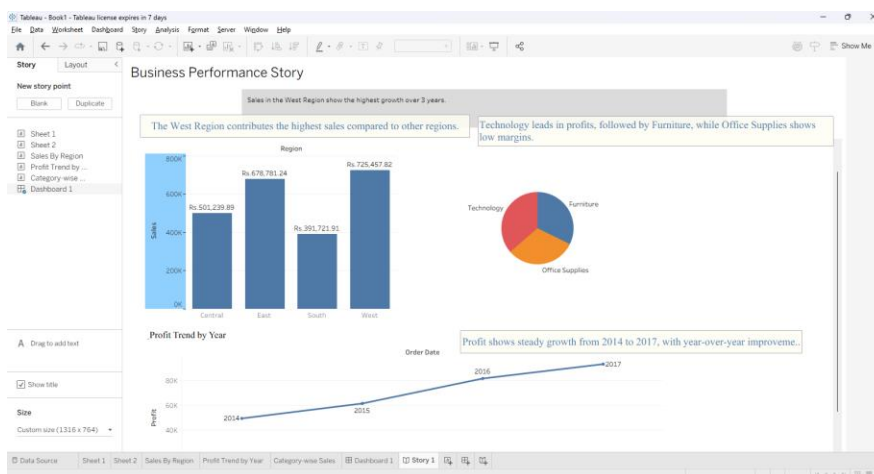
Example: Add Region filter to control all charts.



Step 5. Create a Story

1. From the menu → Story → New Story.
2. A blank Story workspace opens with a “story point” placeholder.

3. Drag a Dashboard or a Worksheet into the story point.
4. Add captions or annotations to explain the insight.
 1. Example: “Sales in the West Region show the highest growth over 3 years.”
5. Add multiple Story Points to narrate data step by step.
 1. Point 1: Sales by Region
 2. Point 2: Profit Trend
 3. Point 3: Category Analysis
6. Format the story → adjust size, fonts, and navigation buttons.



Step 6. Save and Present

1. Save workbook as **.twb** / **.twbx**.
2. Present story → click through story points like slides.

